Tests on building instance trackers on object detection results

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Situtation overview

This document is a summary of tests run on the instance tracker built for the l3f browser app `src/main.js -> InsDetTracker`. We aim to see how well the current tracker is able to handle known problem situations such as

- **size variance between detections** of same real-world object (attempted solution: using the calculated center points as the authoritative location data rather than the bbox info as outputted by the detector directly)
- individual real-world object **may not be detected over every frame** while actually being in the image & **occasional ghost-detections** (detection registered by detector matching no actual real world object); (attempted solution: adding 'TENTATIVE' & 'TEMPORARILY_MISSING' statuses to tracked object & warmup & missing item logic)
- **"flip-flopping" or "category flipping"**; when a real world object is detected as belonging to different object categories between each frame (attempted solution: keeping category history data for individual tracked objects & taking this into account when predicting future locations of objects)

And whether better solutions can be found during the analysis. Most obviously by optimizing the following config constants:

```
/**
 * Constants governing tracking configuration.
 */
const TRACKER_CONFIG = {
  // Lifecycle thresholds
  CONFIRMATION_FRAMES: 2,     // Required consecutive detections before promotion from TENTATIVE
  MISSING_FRAMES_LIMIT: 5,    // Consecutive missed frames before marking DELETED

  // Matching thresholds (in pixel space)
  LOOSE_MATCH_DISTANCE_PX: 40,   // Standard match threshold for category matches/flip-flops
  TIGHT_MATCH_DISTANCE_PX: 15,   // Strict threshold for mismatched categories without flip-flop histor
  BACKUP_MATCH_DISTANCE_PX: 60   // Velocity-direction alignment fallback threshold
};
```

This analysis is conducted on multiple measurements.

Setup

Tests Each of the tests are done with a sampling rate of 8fps and with a measurement duration of about one minute. The tracker config is the same for every measurement in measurement run 1.

Technical Render this document by running `Rscript -e "rmarkdown::render('tracker_experiment_analyais.Rmd')"` From `browser_app/measurements/`

Libraries:

```r
library(jsonlite)
suppressPackageStartupMessages(library(dplyr)) # <- Don't print warning on import
suppressPackageStartupMessages(library(purrr)) # object masked: jsonlite: flatten
library(tibble)
library(ggplot2)
```

Functions:

```r
# Read one JSON object per line, preserving each record as a list.
read_jsonl <- function(path) {
  lines <- readLines(path, warn = FALSE, encoding = "UTF-8")
  lapply(lines, jsonlite::fromJSON, simplifyVector = FALSE)
}

print_track_timeline <- function(records) {
  # The first record is metadata; remaining records are frames.
  frames <- Filter(
    function(x) identical(x$record_type, "frame"),
    records
  )

  for (frame in frames) {
    timestamp <- frame$tracker_state$timestamp
    tracks <- frame$tracker_state$tracks

    if (length(tracks) == 0L) {
      cat(sprintf("[%s] Active tracks: none\n", timestamp))
      next
    }

    track_labels <- vapply(
      tracks,
      function(track) {
        instance_id <- track$instanceId %||% "?"
        category <- track$currentCategory %||% "unknown"
        status <- track$status %||% "unknown"

        sprintf("%s [%s]: %s", instance_id, category, status)
      },
      character(1)
    )

    cat(sprintf(
      "[%s] Active tracks: %s\n",
      timestamp,
      paste(track_labels, collapse = " | ")
    ))
  }
}

# Helper for providing a fallback when a field is NULL.
`%||%` <- function(x, y) {
  if (is.null(x)) y else x
}
```

```r
# Function to split a string into N-character chunks
chunk_string <- function(s, width = 90) {
  n <- nchar(s)
  if (n <= width) return(s)
  starts <- seq(1, n, by = width)
  ends <- pmin(starts + width - 1, n)
  substring(s, starts, ends)
}



print_category_history_values <- function(data, data_name = deparse(substitute(data))) {
  # Keep only frame records
  frame_records <- Filter(
    function(record) identical(record$record_type, "frame"),
    data
  )

  # Collect categoryHistory from every track in every frame
  histories <- unlist(
    lapply(frame_records, function(record) {
      tracks <- record$tracker_state$tracks

      if (is.null(tracks) || length(tracks) == 0) {
        return(list())
      }

      lapply(tracks, function(track) {
        track$categoryHistory
      })
    }),
    recursive = FALSE
  )

  # Handle the case where no tracks or histories were found
  if (length(histories) == 0) {
    cat(
      "Here are all the different categoryHistory values in tracks ",
      "in frame-records in ", data_name, ":\n\n",
      "(No categoryHistory values found.)\n",
      sep = ""
    )
    return(invisible(NULL))
  }

  # Convert each history to a consistent JSON representation
  history_json <- vapply(histories, function(history) {
    if (is.null(history)) {
      return("null")
    }

    jsonlite::toJSON(
      history,
```

```
      auto_unbox = TRUE,
      null = "null",
      na = "null",
      pretty = FALSE
    ) |>
      as.character()
  }, character(1))

  # Print unique values, preserving their first-seen order
  unique_histories <- unique(history_json)

  cat(
    "Here are all the different categoryHistory values in tracks ",
    "in frame-records in ", data_name, ":\n\n",
    paste(unique_histories, collapse = "\n"),
    "\n",
    sep = ""
  )

  invisible(unique_histories)
}

# Function for producing (timestamp, hasBeenConfirmed-count) pairs over a tX
confirmed_tracks_over_time <- function(data) {
  keep <- which(vapply(
    data,
    function(record) identical(record$record_type, "frame"),
    logical(1)
  ))

  map_dfr(keep, function(i) {
    record <- data[[i]]
    tracks <- record$tracker_state$tracks

    confirmed_count <- if (is.null(tracks) || length(tracks) == 0) {
      0L
    } else {
      sum(vapply(
        tracks,
        function(track) isTRUE(track$hasBeenConfirmed),
        logical(1)
      ))
    }

    tibble(
      timestamp_ms = record$timestamp_ms,
      confirmed_tracks = confirmed_count
    )
  })
}

plot_track_trajectories <- function(records, width = 640, height = 480) {
```

```r
points <- map2_dfr(records, seq_along(records), function(record, i) {
  tracks <- record$tracker_state$tracks

  if (is.null(tracks) || length(tracks) == 0) {
    return(tibble())
  }

  map_dfr(tracks, function(track) {
    center <- track$center

    if (is.null(center) ||
        is.null(center$centerX) ||
        is.null(center$centerY)) {
      return(tibble())
    }

    tibble(
      frame = i,
      timestamp_ms = as.numeric(record$timestamp_ms),
      track_id = as.character(track$instanceId),
      x = as.numeric(center$centerX),
      y = as.numeric(center$centerY)
    )
  })
})

if (nrow(points) == 0) {
  stop("No track centers found in the supplied records.")
}

points <- points %>%
  distinct(track_id, x, y, .keep_all = TRUE) %>%
  arrange(track_id, frame)

# Create a segment from each center to the next center of that track.
segments <- points %>%
  group_by(track_id) %>%
  mutate(
    xend = lead(x),
    yend = lead(y)
  ) %>%
  filter(!is.na(xend), !is.na(yend)) %>%
  ungroup()

ggplot() +
  geom_segment(
    data = segments,
    aes(
      x = x, y = y,
      xend = xend, yend = yend,
      color = track_id
    ),
    arrow = grid::arrow(
```

```
        length = grid::unit(0.1, "inches"),
        type = "closed"
      ),
      linewidth = 0.8,
      lineend = "round"
    ) +
    geom_point(
      data = points,
      aes(x = x, y = y, color = track_id),
      size = 2.5
    ) +
    scale_x_continuous(
      position = "top",
      limits = c(0, width),
      breaks = seq(0, width, by = 80),
      expand = c(0, 0)
    ) +
    scale_y_reverse(
      limits = c(height, 0),
      breaks = seq(0, height, by = 60),
      expand = c(0, 0)
    ) +
    coord_fixed() +
    labs(
      title = "Track trajectories",
      x = "X coordinate (pixels)",
      y = "Y coordinate (pixels)",
      color = "Track ID"
    ) +
    theme_minimal(base_size = 12) +
    theme(
      panel.grid.minor = element_blank(),
      legend.position = "right"
    )
}
```

Files in:

```
t1 <- read_jsonl("tracking_1_single_object_stationary.jsonl")
t2 <- read_jsonl("tracking_2_multiple_objects_stationary.jsonl")
t3 <- read_jsonl("tracking_3_single_object_slowly_moving.jsonl")
t4 <- read_jsonl("tracking_4_single_object_rapidly_moving.jsonl")
t5 <- read_jsonl("tracking_5_single_object_enter_frame.jsonl")
t6 <- read_jsonl("tracking_6_single_object_leave_frame.jsonl")
t7 <- read_jsonl("tracking_7_multiple_objects_slowly_moving.jsonl")
```

Data exploration

On the dataformat in general Tracking results are saved in the following format:

- .jsonl
- First line is per-measurement metadata
- Rest of the lines are `(detection, state)`-pairs, where
  - `detection` is neural network output generated by feeding our model a frame sample from the webcam stream

– `state` is the internal state representation generated by feeding `detection` to `InsDetTracker.updateState()` where the state has already consumed the previous history, so $State_n = State_{n-1}.updateState(detection, timestamp)$

To showcase this more clearly; Here are concrete examples of the contents of `t2`:

First 5 lines:

```
cat(paste(unlist(lapply(seq_along(head(readLines("tracking_2_multiple_objects_stationary.jsonl", warn =
  chunks <- chunk_string(head(readLines("tracking_2_multiple_objects_stationary.jsonl", warn = FALSE, e
  c(paste0("--- Line ", i, " ---"), chunks)
})), collapse = "\n"), "\n")
```

```
## --- Line 1 ---
## {"record_type":"metadata","schema_version":1,"experiment_id":"tracking_20261013T153332Z_17
## f204a2","scenario":"Multiple trackable objects are kept stationary in front of the camera"
## ,"custom_notes":"","tracker_config":{"CONFIRMATION_FRAMES":2,"MISSING_FRAMES_LIMIT":5,"LOO
## SE_MATCH_DISTANCE_PX":40,"TIGHT_MATCH_DISTANCE_PX":15,"BACKUP_MATCH_DISTANCE_PX":60},"targ
## et_sampling_fps":8,"start_time_utc":"2026-10-13T15:33:32.279Z","end_time_utc":"2026-10-13T
## 15:34:35.284Z","frame_count":503}
## --- Line 2 ---
## {"record_type":"frame","frame_index":0,"timestamp_ms":511614,"detections":{"detections":[{
## "categories":[{"score":0.3513655364513397,"index":-1,"categoryName":"knife","displayName":
## ""}],"keypoints":[],"boundingBox":{"originX":60,"originY":206,"width":575,"height":266,"an
## gle":0}}]},"tracker_state":{"timestamp":511614,"tracks":[{"instanceId":"#1","status":"TENT
## ATIVE","center":{"centerX":347.5,"centerY":339},"boundingBox":{"originX":60,"originY":206,
## "width":575,"height":266,"angle":0},"velocity":{"vx":0,"vy":0},"history":[{"timestamp":511
## 614,"category":"knife","detectionConfidence":0.3513655364513397,"boundingBox":{"originX":6
## 0,"originY":206,"width":575,"height":266,"angle":0},"center":{"centerX":347.5,"centerY":33
## 9}}],"categoryHistory":{},"currentCategory":"knife","lastDetectionConfidence":0.3513655364
## 513397,"hasBeenConfirmed":false,"createdAt":511614,"updatedAt":511614,"consecutiveHits":1,
## "consecutiveMisses":0}]}}
## --- Line 3 ---
## {"record_type":"frame","frame_index":1,"timestamp_ms":511737,"detections":{"detections":[]
## },"tracker_state":{"timestamp":511737,"tracks":[{"instanceId":"#1","status":"TEMPORARILY_M
## ISSING","center":{"centerX":347.5,"centerY":339},"boundingBox":{"originX":60,"originY":206
## ,"width":575,"height":266,"angle":0},"velocity":{"vx":0,"vy":0},"history":[{"timestamp":51
## 1614,"category":"knife","detectionConfidence":0.3513655364513397,"boundingBox":{"originX":
## 60,"originY":206,"width":575,"height":266,"angle":0},"center":{"centerX":347.5,"centerY":3
## 39}}],"categoryHistory":{},"currentCategory":"knife","lastDetectionConfidence":0.351365536
## 4513397,"hasBeenConfirmed":false,"createdAt":511614,"updatedAt":511614,"consecutiveHits":0
## ,"consecutiveMisses":1}]}}
## --- Line 4 ---
## {"record_type":"frame","frame_index":2,"timestamp_ms":511861,"detections":{"detections":[{
## "categories":[{"score":0.36325934529304504,"index":-1,"categoryName":"banana","displayName
## ":""}],"keypoints":[],"boundingBox":{"originX":6,"originY":231,"width":320,"height":228,"a
## ngle":0}}]},"tracker_state":{"timestamp":511861,"tracks":[{"instanceId":"#1","status":"TEM
## PORARILY_MISSING","center":{"centerX":347.5,"centerY":339},"boundingBox":{"originX":60,"or
## iginY":206,"width":575,"height":266,"angle":0},"velocity":{"vx":0,"vy":0},"history":[{"tim
## estamp":511614,"category":"knife","detectionConfidence":0.3513655364513397,"boundingBox":{
## "originX":60,"originY":206,"width":575,"height":266,"angle":0},"center":{"centerX":347.5,"
## centerY":339}}],"categoryHistory":{},"currentCategory":"knife","lastDetectionConfidence":0
## .3513655364513397,"hasBeenConfirmed":false,"createdAt":511614,"updatedAt":511614,"consecut
## iveHits":0,"consecutiveMisses":2},{"instanceId":"#2","status":"TENTATIVE","center":{"cente
## rX":166,"centerY":345},"boundingBox":{"originX":6,"originY":231,"width":320,"height":228,"
## angle":0},"velocity":{"vx":0,"vy":0},"history":[{"timestamp":511861,"category":"banana","d
## etectionConfidence":0.36325934529304504,"boundingBox":{"originX":6,"originY":231,"width":3
```

```
## 20,"height":228,"angle":0},"center":{"centerX":166,"centerY":345}}],"categoryHistory":{},"
## currentCategory":"banana","lastDetectionConfidence":0.36325934529304504,"hasBeenConfirmed"
## :false,"createdAt":511861,"updatedAt":511861,"consecutiveHits":1,"consecutiveMisses":0}]}}
## --- Line 5 ---
## {"record_type":"frame","frame_index":3,"timestamp_ms":511987,"detections":{"detections":[{
## "categories":[{"score":0.3978668451309204,"index":-1,"categoryName":"banana","displayName"
## :""}],"keypoints":[],"boundingBox":{"originX":6,"originY":245,"width":315,"height":208,"an
## gle":0}}]},"tracker_state":{"timestamp":511987,"tracks":[{"instanceId":"#1","status":"TEMP
## ORARILY_MISSING","center":{"centerX":347.5,"centerY":339},"boundingBox":{"originX":60,"ori
## ginY":206,"width":575,"height":266,"angle":0},"velocity":{"vx":0,"vy":0},"history":[{"time
## stamp":511614,"category":"knife","detectionConfidence":0.3513655364513397,"boundingBox":{"
## originX":60,"originY":206,"width":575,"height":266,"angle":0},"center":{"centerX":347.5,"c
## enterY":339}}],"categoryHistory":{},"currentCategory":"knife","lastDetectionConfidence":0.
## 3513655364513397,"hasBeenConfirmed":false,"createdAt":511614,"updatedAt":511614,"consecuti
## veHits":0,"consecutiveMisses":3},{"instanceId":"#2","status":"CONFIRMED","center":{"center
## X":163.5,"centerY":349},"boundingBox":{"originX":6,"originY":245,"width":315,"height":208,
## "angle":0},"velocity":{"vx":-0.01984126984126984,"vy":0.031746031746031744},"history":[{"t
## imestamp":511861,"category":"banana","detectionConfidence":0.36325934529304504,"boundingBo
## x":{"originX":6,"originY":231,"width":320,"height":228,"angle":0},"center":{"centerX":166,
## "centerY":345}},{"timestamp":511987,"category":"banana","detectionConfidence":0.3978668451
## 309204,"boundingBox":{"originX":6,"originY":245,"width":315,"height":208,"angle":0},"cente
## r":{"centerX":163.5,"centerY":349}}],"categoryHistory":{},"currentCategory":"banana","last
## DetectionConfidence":0.3978668451309204,"hasBeenConfirmed":true,"createdAt":511861,"update
## dAt":511987,"consecutiveHits":2,"consecutiveMisses":0}]}}
```

Last 10 lines:

```
cat(paste(unlist(lapply(seq_along(tail(readLines("tracking_2_multiple_objects_stationary.jsonl", warn =
  chunks <- chunk_string(tail(readLines("tracking_2_multiple_objects_stationary.jsonl", warn = FALSE, e
  c(paste0("--- Line ", i, " ---"), chunks)
})), collapse = "\n"), "\n")
```

```
## --- Line 1 ---
## {"record_type":"frame","frame_index":493,"timestamp_ms":573285,"detections":{"detections":
## []},"tracker_state":{"timestamp":573285,"tracks":[{"instanceId":"#29","status":"TEMPORARIL
## Y_MISSING","center":{"centerX":448,"centerY":247.5},"boundingBox":{"originX":400,"originY"
## :198,"width":96,"height":99,"angle":0},"velocity":{"vx":0.009000000000000001,"vy":0.003976
## 190476190476},"history":[{"timestamp":572285,"category":"apple","detectionConfidence":0.42
## 371854186058044,"boundingBox":{"originX":398,"originY":197,"width":91,"height":97,"angle":
## 0},"center":{"centerX":443.5,"centerY":245.5}},{"timestamp":572410,"category":"apple","det
## ectionConfidence":0.5022628307342529,"boundingBox":{"originX":398,"originY":198,"width":92
## ,"height":96,"angle":0},"center":{"centerX":444,"centerY":246}},{"timestamp":572536,"categ
## ory":"apple","detectionConfidence":0.43481603264808655,"boundingBox":{"originX":397,"origi
## nY":198,"width":94,"height":99,"angle":0},"center":{"centerX":444,"centerY":247.5}},{"time
## stamp":572661,"category":"apple","detectionConfidence":0.46943068504333496,"boundingBox":{
## "originX":399,"originY":203,"width":93,"height":94,"angle":0},"center":{"centerX":445.5,"c
## enterY":250}},{"timestamp":572786,"category":"apple","detectionConfidence":0.4329078793525
## 696,"boundingBox":{"originX":400,"originY":198,"width":96,"height":99,"angle":0},"center":
## {"centerX":448,"centerY":247.5}}],"categoryHistory":{},"currentCategory":"apple","lastDete
## ctionConfidence":0.4329078793525696,"hasBeenConfirmed":true,"createdAt":571031,"updatedAt"
## :572786,"consecutiveHits":0,"consecutiveMisses":4},{"instanceId":"#30","status":"TEMPORARI
## LY_MISSING","center":{"centerX":517.5,"centerY":337},"boundingBox":{"originX":398,"originY
## ":197,"width":239,"height":280,"angle":0},"velocity":{"vx":0,"vy":0},"history":[{"timestam
## p":572786,"category":"banana","detectionConfidence":0.4471163749694824,"boundingBox":{"ori
## ginX":398,"originY":197,"width":239,"height":280,"angle":0},"center":{"centerX":517.5,"cen
```

```
## terY":337}}],"categoryHistory":{},"currentCategory":"banana","lastDetectionConfidence":0.4
## 471163749694824,"hasBeenConfirmed":false,"createdAt":572786,"updatedAt":572786,"consecutiv
## eHits":0,"consecutiveMisses":4}]}}
## --- Line 2 ---
## {"record_type":"frame","frame_index":494,"timestamp_ms":573410,"detections":{"detections":
## []},"tracker_state":{"timestamp":573410,"tracks":[]}}
## --- Line 3 ---
## {"record_type":"frame","frame_index":495,"timestamp_ms":573535,"detections":{"detections":
## []},"tracker_state":{"timestamp":573535,"tracks":[]}}
## --- Line 4 ---
## {"record_type":"frame","frame_index":496,"timestamp_ms":573661,"detections":{"detections":
## []},"tracker_state":{"timestamp":573661,"tracks":[]}}
## --- Line 5 ---
## {"record_type":"frame","frame_index":497,"timestamp_ms":573787,"detections":{"detections":
## []},"tracker_state":{"timestamp":573787,"tracks":[]}}
## --- Line 6 ---
## {"record_type":"frame","frame_index":498,"timestamp_ms":573911,"detections":{"detections":
## []},"tracker_state":{"timestamp":573911,"tracks":[]}}
## --- Line 7 ---
## {"record_type":"frame","frame_index":499,"timestamp_ms":574037,"detections":{"detections":
## []},"tracker_state":{"timestamp":574037,"tracks":[]}}
## --- Line 8 ---
## {"record_type":"frame","frame_index":500,"timestamp_ms":574161,"detections":{"detections":
## []},"tracker_state":{"timestamp":574161,"tracks":[]}}
## --- Line 9 ---
## {"record_type":"frame","frame_index":501,"timestamp_ms":574286,"detections":{"detections":
## []},"tracker_state":{"timestamp":574286,"tracks":[]}}
## --- Line 10 ---
## {"record_type":"frame","frame_index":502,"timestamp_ms":574410,"detections":{"detections":
## []},"tracker_state":{"timestamp":574410,"tracks":[]}}
```

Frame 200 pretty-printed:

```
cat(jsonlite::prettify(readLines("tracking_2_multiple_objects_stationary.jsonl", warn = FALSE, encoding
```

```
## {
##     "record_type": "frame",
##     "frame_index": 200,
##     "timestamp_ms": 536633,
##     "detections": {
##         "detections": [
##
##         ]
##     },
##     "tracker_state": {
##         "timestamp": 536633,
##         "tracks": [
##             {
##                 "instanceId": "#15",
##                 "status": "TEMPORARILY_MISSING",
##                 "center": {
##                     "centerX": 267,
##                     "centerY": 306.5
##                 },
##                 "boundingBox": {
```

```
##                     "originX": 146,
##                     "originY": 253,
##                     "width": 242,
##                     "height": 107,
##                     "angle": 0
##                 },
##                 "velocity": {
##                     "vx": 0.01000064004096262,
##                     "vy": 0.002048131080389145
##                 },
##                 "history": [
##                     {
##                         "timestamp": 535633,
##                         "category": "banana",
##                         "detectionConfidence": 0.44796305894851685,
##                         "boundingBox": {
##                             "originX": 137,
##                             "originY": 252,
##                             "width": 250,
##                             "height": 107,
##                             "angle": 0
##                         },
##                         "center": {
##                             "centerX": 262,
##                             "centerY": 305.5
##                         }
##                     },
##                     {
##                         "timestamp": 535759,
##                         "category": "banana",
##                         "detectionConfidence": 0.4665168821811676,
##                         "boundingBox": {
##                             "originX": 141,
##                             "originY": 252,
##                             "width": 253,
##                             "height": 107,
##                             "angle": 0
##                         },
##                         "center": {
##                             "centerX": 267.5,
##                             "centerY": 305.5
##                         }
##                     },
##                     {
##                         "timestamp": 535883,
##                         "category": "banana",
##                         "detectionConfidence": 0.4576960802078247,
##                         "boundingBox": {
##                             "originX": 144,
##                             "originY": 253,
##                             "width": 253,
##                             "height": 108,
##                             "angle": 0
##                         },
```

```
##                         "center": {
##                             "centerX": 270.5,
##                             "centerY": 307
##                         }
##                     },
##                     {
##                         "timestamp": 536009,
##                         "category": "banana",
##                         "detectionConfidence": 0.40532782673835754,
##                         "boundingBox": {
##                             "originX": 146,
##                             "originY": 253,
##                             "width": 243,
##                             "height": 106,
##                             "angle": 0
##                         },
##                         "center": {
##                             "centerX": 267.5,
##                             "centerY": 306
##                         }
##                     },
##                     {
##                         "timestamp": 536133,
##                         "category": "banana",
##                         "detectionConfidence": 0.353581041097641,
##                         "boundingBox": {
##                             "originX": 146,
##                             "originY": 253,
##                             "width": 242,
##                             "height": 107,
##                             "angle": 0
##                         },
##                         "center": {
##                             "centerX": 267,
##                             "centerY": 306.5
##                         }
##                     }
##                 ],
##                 "categoryHistory": {
##
##                 },
##                 "currentCategory": "banana",
##                 "lastDetectionConfidence": 0.353581041097641,
##                 "hasBeenConfirmed": true,
##                 "createdAt": 534883,
##                 "updatedAt": 536133,
##                 "consecutiveHits": 0,
##                 "consecutiveMisses": 4
##             },
##             {
##                 "instanceId": "#16",
##                 "status": "TEMPORARILY_MISSING",
##                 "center": {
##                     "centerX": 420.5,
```

```
##                     "centerY": 252.5
##                 },
##                 "boundingBox": {
##                     "originX": 376,
##                     "originY": 205,
##                     "width": 89,
##                     "height": 95,
##                     "angle": 0
##                 },
##                 "velocity": {
##                     "vx": 0,
##                     "vy": 0
##                 },
##                 "history": [
##                     {
##                         "timestamp": 536259,
##                         "category": "apple",
##                         "detectionConfidence": 0.3533577024936676,
##                         "boundingBox": {
##                             "originX": 376,
##                             "originY": 205,
##                             "width": 89,
##                             "height": 95,
##                             "angle": 0
##                         },
##                         "center": {
##                             "centerX": 420.5,
##                             "centerY": 252.5
##                         }
##                     }
##                 ],
##                 "categoryHistory": {
##
##                 },
##                 "currentCategory": "apple",
##                 "lastDetectionConfidence": 0.3533577024936676,
##                 "hasBeenConfirmed": false,
##                 "createdAt": 536259,
##                 "updatedAt": 536259,
##                 "consecutiveHits": 0,
##                 "consecutiveMisses": 3
##             }
##         ]
##     }
## }
##
```

On measurement set 1 Experiment metadata:

```
cat(jsonlite::prettify(readLines("tracking_1_single_object_stationary.jsonl", warn = FALSE, encoding =
```

```
## {
##     "record_type": "metadata",
##     "schema_version": 1,
##     "experiment_id": "tracking_20261013T152343Z_6a48668d",
```

```
##     "scenario": "One trackable object is kept stationary in front of the camera",
##     "custom_notes": "",
##     "tracker_config": {
##         "CONFIRMATION_FRAMES": 2,
##         "MISSING_FRAMES_LIMIT": 5,
##         "LOOSE_MATCH_DISTANCE_PX": 40,
##         "TIGHT_MATCH_DISTANCE_PX": 15,
##         "BACKUP_MATCH_DISTANCE_PX": 60
##     },
##     "target_sampling_fps": 8,
##     "start_time_utc": "2026-10-13T15:23:43.854Z",
##     "end_time_utc": "2026-10-13T15:24:54.615Z",
##     "frame_count": 482
## }
##
```

Track timeline:

```
print_track_timeline(t1)
```

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## [1716028] Active tracks: #1 [bowl]: CONFIRMED
## [1716153] Active tracks: #1 [bowl]: CONFIRMED
## [1716278] Active tracks: #1 [bowl]: CONFIRMED
## [1716403] Active tracks: #1 [bowl]: CONFIRMED
## [1716529] Active tracks: #1 [bowl]: CONFIRMED
```

```
## [1716653] Active tracks: #1 [bowl]: CONFIRMED
## [1716779] Active tracks: #1 [bowl]: CONFIRMED
## [1716904] Active tracks: #1 [bowl]: CONFIRMED
## [1717030] Active tracks: #1 [bowl]: CONFIRMED
## [1717154] Active tracks: #1 [bowl]: CONFIRMED
## [1717280] Active tracks: #1 [bowl]: CONFIRMED
## [1717405] Active tracks: #1 [bowl]: CONFIRMED
## [1717529] Active tracks: #1 [bowl]: CONFIRMED
## [1717654] Active tracks: #1 [bowl]: CONFIRMED
## [1717780] Active tracks: #1 [bowl]: CONFIRMED
## [1717904] Active tracks: #1 [bowl]: CONFIRMED
## [1718029] Active tracks: #1 [bowl]: CONFIRMED
## [1718155] Active tracks: #1 [bowl]: CONFIRMED
## [1718280] Active tracks: #1 [bowl]: CONFIRMED
## [1718404] Active tracks: #1 [bowl]: CONFIRMED
## [1718530] Active tracks: #1 [bowl]: CONFIRMED
## [1718655] Active tracks: #1 [bowl]: CONFIRMED
## [1718780] Active tracks: #1 [bowl]: CONFIRMED
## [1718905] Active tracks: #1 [bowl]: CONFIRMED
## [1719030] Active tracks: #1 [bowl]: CONFIRMED
## [1719156] Active tracks: #1 [bowl]: CONFIRMED
## [1719282] Active tracks: #1 [bowl]: CONFIRMED
## [1719406] Active tracks: #1 [bowl]: CONFIRMED
## [1719531] Active tracks: #1 [bowl]: CONFIRMED
## [1719656] Active tracks: #1 [bowl]: CONFIRMED
## [1719781] Active tracks: #1 [bowl]: CONFIRMED
## [1719906] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1720031] Active tracks: #1 [bowl]: CONFIRMED
## [1720157] Active tracks: #1 [bowl]: CONFIRMED
## [1720282] Active tracks: #1 [bowl]: CONFIRMED
## [1720407] Active tracks: #1 [bowl]: CONFIRMED
## [1720532] Active tracks: #1 [bowl]: CONFIRMED
## [1720657] Active tracks: #1 [bowl]: CONFIRMED
## [1720782] Active tracks: #1 [bowl]: CONFIRMED
## [1720906] Active tracks: #1 [bowl]: CONFIRMED
## [1721032] Active tracks: #1 [bowl]: CONFIRMED
## [1721159] Active tracks: #1 [bowl]: CONFIRMED
## [1721283] Active tracks: #1 [bowl]: CONFIRMED
## [1721409] Active tracks: #1 [bowl]: CONFIRMED
## [1721532] Active tracks: #1 [bowl]: CONFIRMED
## [1721658] Active tracks: #1 [bowl]: CONFIRMED
## [1721783] Active tracks: #1 [bowl]: CONFIRMED
## [1721907] Active tracks: #1 [bowl]: CONFIRMED
## [1722033] Active tracks: #1 [bowl]: CONFIRMED
## [1722159] Active tracks: #1 [bowl]: CONFIRMED
## [1722283] Active tracks: #1 [bowl]: CONFIRMED
## [1722407] Active tracks: #1 [bowl]: CONFIRMED
## [1722533] Active tracks: #1 [bowl]: CONFIRMED
## [1722658] Active tracks: #1 [bowl]: CONFIRMED
## [1722783] Active tracks: #1 [bowl]: CONFIRMED
## [1722908] Active tracks: #1 [bowl]: CONFIRMED
## [1723034] Active tracks: #1 [bowl]: CONFIRMED
## [1723158] Active tracks: #1 [bowl]: CONFIRMED
## [1723283] Active tracks: #1 [bowl]: CONFIRMED
```

```
## [1723409] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1723534] Active tracks: #1 [bowl]: CONFIRMED
## [1723659] Active tracks: #1 [bowl]: CONFIRMED
## [1723784] Active tracks: #1 [bowl]: CONFIRMED
## [1723909] Active tracks: #1 [bowl]: CONFIRMED
## [1724034] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1724159] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1724284] Active tracks: #1 [bowl]: CONFIRMED
## [1724409] Active tracks: #1 [bowl]: CONFIRMED
## [1724534] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1724660] Active tracks: #1 [bowl]: CONFIRMED
## [1724785] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1724910] Active tracks: #1 [bowl]: CONFIRMED
## [1725035] Active tracks: #1 [bowl]: CONFIRMED
## [1725162] Active tracks: #1 [bowl]: CONFIRMED
## [1725285] Active tracks: #1 [bowl]: CONFIRMED
## [1725409] Active tracks: #1 [bowl]: CONFIRMED
## [1725535] Active tracks: #1 [bowl]: CONFIRMED
## [1725660] Active tracks: #1 [bowl]: CONFIRMED
## [1725785] Active tracks: #1 [bowl]: CONFIRMED
## [1725910] Active tracks: #1 [bowl]: CONFIRMED
## [1726035] Active tracks: #1 [bowl]: CONFIRMED
## [1726160] Active tracks: #1 [bowl]: CONFIRMED
## [1726285] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1726410] Active tracks: #1 [bowl]: CONFIRMED
## [1726536] Active tracks: #1 [bowl]: CONFIRMED
## [1726660] Active tracks: #1 [bowl]: CONFIRMED
## [1726786] Active tracks: #1 [bowl]: CONFIRMED
## [1726911] Active tracks: #1 [bowl]: CONFIRMED
## [1727036] Active tracks: #1 [bowl]: CONFIRMED
## [1727161] Active tracks: #1 [bowl]: CONFIRMED
## [1727286] Active tracks: #1 [bowl]: CONFIRMED
## [1727411] Active tracks: #1 [bowl]: CONFIRMED
## [1727536] Active tracks: #1 [bowl]: CONFIRMED
## [1727661] Active tracks: #1 [bowl]: CONFIRMED
## [1727786] Active tracks: #1 [bowl]: CONFIRMED
## [1727912] Active tracks: #1 [bowl]: CONFIRMED
## [1728037] Active tracks: #1 [bowl]: CONFIRMED
## [1728161] Active tracks: #1 [bowl]: CONFIRMED
## [1728287] Active tracks: #1 [bowl]: CONFIRMED
## [1728411] Active tracks: #1 [bowl]: CONFIRMED
## [1728537] Active tracks: #1 [bowl]: CONFIRMED
## [1728662] Active tracks: #1 [bowl]: CONFIRMED
## [1728787] Active tracks: #1 [bowl]: CONFIRMED
## [1728912] Active tracks: #1 [bowl]: CONFIRMED
## [1729037] Active tracks: #1 [bowl]: CONFIRMED
## [1729162] Active tracks: #1 [bowl]: CONFIRMED
## [1729288] Active tracks: #1 [bowl]: CONFIRMED
## [1729413] Active tracks: #1 [bowl]: CONFIRMED
## [1729537] Active tracks: #1 [bowl]: CONFIRMED
## [1729662] Active tracks: #1 [bowl]: CONFIRMED
## [1729788] Active tracks: #1 [bowl]: CONFIRMED
## [1729913] Active tracks: #1 [bowl]: CONFIRMED
## [1730038] Active tracks: #1 [bowl]: CONFIRMED
```

```
## [1730163] Active tracks: #1 [bowl]: CONFIRMED
## [1730288] Active tracks: #1 [bowl]: CONFIRMED
## [1730413] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1730540] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1730665] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1730789] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1730915] Active tracks: none
## [1731039] Active tracks: none
## [1731166] Active tracks: none
## [1731291] Active tracks: #2 [bowl]: TENTATIVE
## [1731414] Active tracks: #2 [bowl]: TEMPORARILY_MISSING
## [1731539] Active tracks: #2 [bowl]: TEMPORARILY_MISSING
## [1731664] Active tracks: #2 [bowl]: TENTATIVE
## [1731790] Active tracks: #2 [bowl]: TEMPORARILY_MISSING
## [1731914] Active tracks: #2 [bowl]: TEMPORARILY_MISSING
```

On measurement set 2 Experiment metadata:

```
cat(jsonlite::prettify(readLines("tracking_2_multiple_objects_stationary.jsonl", warn = FALSE, encoding
```

```
## {
##     "record_type": "metadata",
##     "schema_version": 1,
##     "experiment_id": "tracking_20261013T153332Z_17f204a2",
##     "scenario": "Multiple trackable objects are kept stationary in front of the camera",
##     "custom_notes": "",
##     "tracker_config": {
##         "CONFIRMATION_FRAMES": 2,
##         "MISSING_FRAMES_LIMIT": 5,
##         "LOOSE_MATCH_DISTANCE_PX": 40,
##         "TIGHT_MATCH_DISTANCE_PX": 15,
##         "BACKUP_MATCH_DISTANCE_PX": 60
##     },
##     "target_sampling_fps": 8,
##     "start_time_utc": "2026-10-13T15:33:32.279Z",
##     "end_time_utc": "2026-10-13T15:34:35.284Z",
##     "frame_count": 503
## }
##
```

Track timeline:

```
print_track_timeline(t2)
```

```
## [511614] Active tracks: #1 [knife]: TENTATIVE
## [511737] Active tracks: #1 [knife]: TEMPORARILY_MISSING
## [511861] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #2 [banana]: TENTATIVE
## [511987] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #2 [banana]: CONFIRMED
## [512111] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #2 [banana]: CONFIRMED
## [512238] Active tracks: #2 [banana]: TEMPORARILY_MISSING
## [512361] Active tracks: #2 [banana]: TEMPORARILY_MISSING
## [512488] Active tracks: #2 [banana]: CONFIRMED
## [512611] Active tracks: #2 [banana]: CONFIRMED
## [512737] Active tracks: #2 [banana]: CONFIRMED
## [512862] Active tracks: #2 [banana]: TEMPORARILY_MISSING
## [512987] Active tracks: #2 [banana]: TEMPORARILY_MISSING
```

```
## [513112] Active tracks: #2 [spoon]: CONFIRMED
## [513237] Active tracks: #2 [spoon]: CONFIRMED
## [513362] Active tracks: #2 [spoon]: CONFIRMED | #3 [apple]: TENTATIVE
## [513492] Active tracks: #2 [spoon]: CONFIRMED | #3 [apple]: CONFIRMED
## [513618] Active tracks: #2 [spoon]: CONFIRMED | #3 [apple]: CONFIRMED
## [513743] Active tracks: #2 [spoon]: CONFIRMED | #3 [apple]: CONFIRMED
## [513868] Active tracks: #2 [spoon]: CONFIRMED | #3 [apple]: CONFIRMED
## [513994] Active tracks: #2 [spoon]: TEMPORARILY_MISSING | #3 [apple]: CONFIRMED
## [514118] Active tracks: #2 [spoon]: CONFIRMED | #3 [apple]: CONFIRMED
## [514244] Active tracks: #2 [spoon]: TEMPORARILY_MISSING | #3 [apple]: CONFIRMED
## [514368] Active tracks: #2 [spoon]: TEMPORARILY_MISSING | #3 [apple]: CONFIRMED
## [514495] Active tracks: #2 [spoon]: TEMPORARILY_MISSING | #3 [apple]: CONFIRMED
## [514618] Active tracks: #2 [spoon]: CONFIRMED | #3 [apple]: CONFIRMED
## [514744] Active tracks: #2 [spoon]: TEMPORARILY_MISSING | #3 [apple]: CONFIRMED
## [514869] Active tracks: #2 [spoon]: TEMPORARILY_MISSING | #3 [apple]: CONFIRMED
## [514995] Active tracks: #2 [spoon]: TEMPORARILY_MISSING | #3 [apple]: CONFIRMED
## [515119] Active tracks: #2 [spoon]: TEMPORARILY_MISSING | #3 [apple]: CONFIRMED
## [515245] Active tracks: #3 [apple]: CONFIRMED
## [515369] Active tracks: #3 [apple]: CONFIRMED
## [515496] Active tracks: #3 [apple]: CONFIRMED
## [515620] Active tracks: #3 [apple]: CONFIRMED
## [515746] Active tracks: #3 [apple]: CONFIRMED
## [515870] Active tracks: #3 [apple]: CONFIRMED
## [515995] Active tracks: #3 [apple]: CONFIRMED
## [516119] Active tracks: #3 [apple]: CONFIRMED
## [516246] Active tracks: #3 [apple]: CONFIRMED | #4 [banana]: TENTATIVE
## [516370] Active tracks: #3 [apple]: CONFIRMED | #4 [spoon]: CONFIRMED
## [516495] Active tracks: #3 [apple]: CONFIRMED | #4 [spoon]: TEMPORARILY_MISSING
## [516620] Active tracks: #3 [apple]: CONFIRMED | #4 [spoon]: TEMPORARILY_MISSING | #5 [knife]: TENTATI
## [516746] Active tracks: #3 [apple]: CONFIRMED | #4 [spoon]: TEMPORARILY_MISSING | #5 [knife]: CONFIRM
## [516870] Active tracks: #3 [apple]: CONFIRMED | #4 [spoon]: TEMPORARILY_MISSING | #5 [knife]: CONFIRM
## [516996] Active tracks: #3 [apple]: CONFIRMED | #5 [knife]: CONFIRMED
## [517119] Active tracks: #3 [apple]: CONFIRMED | #5 [knife]: CONFIRMED
## [517245] Active tracks: #3 [apple]: CONFIRMED | #5 [knife]: CONFIRMED
## [517370] Active tracks: #3 [apple]: TEMPORARILY_MISSING | #5 [knife]: CONFIRMED
## [517494] Active tracks: #3 [apple]: CONFIRMED | #5 [spoon]: CONFIRMED
## [517620] Active tracks: #3 [apple]: CONFIRMED | #5 [knife]: CONFIRMED
## [517746] Active tracks: #3 [apple]: TEMPORARILY_MISSING | #5 [knife]: CONFIRMED
## [517870] Active tracks: #3 [apple]: CONFIRMED | #5 [knife]: CONFIRMED
## [517996] Active tracks: #3 [apple]: TEMPORARILY_MISSING | #5 [spoon]: CONFIRMED
## [518121] Active tracks: #3 [apple]: CONFIRMED | #5 [spoon]: TEMPORARILY_MISSING
## [518246] Active tracks: #3 [apple]: CONFIRMED | #5 [spoon]: TEMPORARILY_MISSING
## [518371] Active tracks: #3 [apple]: TEMPORARILY_MISSING | #5 [spoon]: CONFIRMED
## [518497] Active tracks: #3 [apple]: CONFIRMED | #5 [spoon]: TEMPORARILY_MISSING
## [518621] Active tracks: #3 [apple]: CONFIRMED | #5 [spoon]: TEMPORARILY_MISSING
## [518747] Active tracks: #3 [apple]: CONFIRMED | #5 [spoon]: TEMPORARILY_MISSING
## [518872] Active tracks: #3 [apple]: CONFIRMED | #5 [spoon]: TEMPORARILY_MISSING
## [518997] Active tracks: #3 [apple]: CONFIRMED
## [519122] Active tracks: #3 [apple]: CONFIRMED
## [519247] Active tracks: #3 [apple]: CONFIRMED
## [519372] Active tracks: #3 [apple]: CONFIRMED
## [519497] Active tracks: #3 [apple]: CONFIRMED
## [519621] Active tracks: #3 [apple]: CONFIRMED
## [519747] Active tracks: #3 [apple]: CONFIRMED
```

```
## [519872] Active tracks: #3 [apple]: CONFIRMED
## [519996] Active tracks: #3 [apple]: CONFIRMED
## [520122] Active tracks: #3 [apple]: CONFIRMED
## [520247] Active tracks: #3 [apple]: CONFIRMED
## [520372] Active tracks: #3 [apple]: CONFIRMED
## [520498] Active tracks: #3 [apple]: CONFIRMED
## [520622] Active tracks: #3 [apple]: CONFIRMED
## [520747] Active tracks: #3 [apple]: CONFIRMED | #6 [banana]: TENTATIVE
## [520873] Active tracks: #3 [apple]: CONFIRMED | #6 [banana]: TEMPORARILY_MISSING
## [520998] Active tracks: #3 [apple]: CONFIRMED | #6 [banana]: TEMPORARILY_MISSING
## [521122] Active tracks: #3 [apple]: CONFIRMED | #6 [banana]: TEMPORARILY_MISSING
## [521247] Active tracks: #3 [apple]: CONFIRMED | #6 [banana]: TEMPORARILY_MISSING
## [521372] Active tracks: #3 [apple]: CONFIRMED
## [521498] Active tracks: #3 [apple]: CONFIRMED
## [521623] Active tracks: #3 [apple]: CONFIRMED
## [521748] Active tracks: #3 [apple]: CONFIRMED
## [521873] Active tracks: #3 [apple]: CONFIRMED
## [521998] Active tracks: #3 [apple]: CONFIRMED
## [522124] Active tracks: #3 [apple]: CONFIRMED
## [522248] Active tracks: #3 [apple]: CONFIRMED
## [522373] Active tracks: #3 [apple]: CONFIRMED
## [522498] Active tracks: #3 [apple]: CONFIRMED | #7 [banana]: TENTATIVE
## [522623] Active tracks: #3 [apple]: CONFIRMED | #7 [banana]: CONFIRMED
## [522748] Active tracks: #3 [apple]: CONFIRMED | #7 [banana]: CONFIRMED
## [522874] Active tracks: #3 [apple]: CONFIRMED | #7 [banana]: CONFIRMED
## [522998] Active tracks: #3 [apple]: CONFIRMED | #7 [banana]: CONFIRMED
## [523124] Active tracks: #3 [apple]: CONFIRMED | #7 [banana]: TEMPORARILY_MISSING
## [523249] Active tracks: #3 [apple]: CONFIRMED | #7 [banana]: TEMPORARILY_MISSING
## [523374] Active tracks: #3 [apple]: CONFIRMED | #7 [banana]: TEMPORARILY_MISSING
## [523499] Active tracks: #3 [apple]: CONFIRMED | #7 [banana]: TEMPORARILY_MISSING
## [523623] Active tracks: #3 [apple]: CONFIRMED
## [523750] Active tracks: #3 [apple]: CONFIRMED
## [523874] Active tracks: #3 [apple]: CONFIRMED | #8 [banana]: TENTATIVE
## [523999] Active tracks: #3 [apple]: CONFIRMED | #8 [banana]: TEMPORARILY_MISSING
## [524124] Active tracks: #3 [apple]: CONFIRMED | #8 [banana]: TEMPORARILY_MISSING
## [524251] Active tracks: #3 [apple]: CONFIRMED | #8 [banana]: TEMPORARILY_MISSING
## [524374] Active tracks: #3 [apple]: CONFIRMED | #8 [banana]: TEMPORARILY_MISSING
## [524499] Active tracks: #3 [apple]: CONFIRMED
## [524625] Active tracks: #3 [apple]: CONFIRMED
## [524751] Active tracks: #3 [apple]: CONFIRMED
## [524875] Active tracks: #3 [apple]: CONFIRMED
## [524999] Active tracks: #3 [apple]: CONFIRMED
## [525125] Active tracks: #3 [apple]: CONFIRMED
## [525250] Active tracks: #3 [apple]: CONFIRMED
## [525375] Active tracks: #3 [apple]: CONFIRMED
## [525500] Active tracks: #3 [apple]: CONFIRMED
## [525625] Active tracks: #3 [apple]: CONFIRMED
## [525751] Active tracks: #3 [apple]: CONFIRMED
## [525875] Active tracks: #3 [apple]: CONFIRMED
## [526000] Active tracks: #3 [apple]: CONFIRMED
## [526126] Active tracks: #3 [apple]: CONFIRMED
## [526253] Active tracks: #3 [apple]: CONFIRMED | #9 [banana]: TENTATIVE
## [526376] Active tracks: #3 [apple]: CONFIRMED | #9 [banana]: CONFIRMED
## [526501] Active tracks: #3 [apple]: CONFIRMED | #9 [banana]: CONFIRMED
```

```
## [526626] Active tracks: #3 [apple]: CONFIRMED | #9 [banana]: CONFIRMED
## [526752] Active tracks: #3 [apple]: TEMPORARILY_MISSING | #9 [banana]: TEMPORARILY_MISSING
## [526876] Active tracks: #3 [apple]: TEMPORARILY_MISSING | #9 [banana]: TEMPORARILY_MISSING
## [527001] Active tracks: #3 [apple]: TEMPORARILY_MISSING | #9 [banana]: TEMPORARILY_MISSING
## [527126] Active tracks: #3 [apple]: TEMPORARILY_MISSING | #9 [banana]: TEMPORARILY_MISSING
## [527252] Active tracks: none
## [527376] Active tracks: none
## [527501] Active tracks: none
## [527626] Active tracks: none
## [527752] Active tracks: none
## [527877] Active tracks: none
## [528002] Active tracks: none
## [528126] Active tracks: none
## [528252] Active tracks: none
## [528377] Active tracks: none
## [528502] Active tracks: none
## [528627] Active tracks: none
## [528752] Active tracks: none
## [528877] Active tracks: none
## [529002] Active tracks: none
## [529127] Active tracks: none
## [529254] Active tracks: none
## [529378] Active tracks: none
## [529505] Active tracks: none
## [529633] Active tracks: none
## [529757] Active tracks: none
## [529880] Active tracks: none
## [530003] Active tracks: none
## [530129] Active tracks: none
## [530254] Active tracks: none
## [530379] Active tracks: none
## [530503] Active tracks: none
## [530630] Active tracks: none
## [530754] Active tracks: none
## [530879] Active tracks: none
## [531004] Active tracks: none
## [531129] Active tracks: none
## [531254] Active tracks: none
## [531380] Active tracks: none
## [531505] Active tracks: none
## [531630] Active tracks: #10 [apple]: TENTATIVE
## [531755] Active tracks: #10 [apple]: CONFIRMED
## [531880] Active tracks: #10 [apple]: CONFIRMED
## [532005] Active tracks: #10 [apple]: CONFIRMED
## [532130] Active tracks: #10 [apple]: CONFIRMED
## [532255] Active tracks: #10 [apple]: TEMPORARILY_MISSING
## [532380] Active tracks: #10 [apple]: TEMPORARILY_MISSING | #11 [knife]: TENTATIVE
## [532506] Active tracks: #10 [apple]: TEMPORARILY_MISSING | #11 [knife]: CONFIRMED
## [532631] Active tracks: #10 [apple]: TEMPORARILY_MISSING | #11 [knife]: CONFIRMED
## [532757] Active tracks: #11 [knife]: CONFIRMED
## [532881] Active tracks: #11 [knife]: TEMPORARILY_MISSING
## [533006] Active tracks: #11 [knife]: CONFIRMED | #12 [apple]: TENTATIVE
## [533131] Active tracks: #11 [knife]: CONFIRMED | #12 [apple]: CONFIRMED
## [533256] Active tracks: #11 [knife]: CONFIRMED | #12 [apple]: TEMPORARILY_MISSING
```

```
## [533381] Active tracks: #11 [knife]: CONFIRMED | #12 [apple]: TEMPORARILY_MISSING
## [533506] Active tracks: #11 [knife]: CONFIRMED | #12 [apple]: TEMPORARILY_MISSING
## [533632] Active tracks: #11 [knife]: TEMPORARILY_MISSING | #12 [apple]: CONFIRMED
## [533756] Active tracks: #11 [knife]: CONFIRMED | #12 [apple]: TEMPORARILY_MISSING
## [533882] Active tracks: #11 [knife]: CONFIRMED | #12 [apple]: TEMPORARILY_MISSING
## [534006] Active tracks: #11 [knife]: TEMPORARILY_MISSING | #12 [apple]: TEMPORARILY_MISSING
## [534132] Active tracks: #11 [knife]: TEMPORARILY_MISSING | #12 [apple]: TEMPORARILY_MISSING | #13 [ba
## [534256] Active tracks: #11 [knife]: TEMPORARILY_MISSING | #13 [banana]: TEMPORARILY_MISSING
## [534383] Active tracks: #11 [knife]: TEMPORARILY_MISSING | #13 [banana]: TEMPORARILY_MISSING | #14 [a
## [534507] Active tracks: #13 [banana]: TEMPORARILY_MISSING | #14 [apple]: CONFIRMED
## [534632] Active tracks: #13 [banana]: TEMPORARILY_MISSING | #14 [apple]: CONFIRMED
## [534758] Active tracks: #14 [apple]: CONFIRMED
## [534883] Active tracks: #14 [apple]: TEMPORARILY_MISSING | #15 [banana]: TENTATIVE
## [535008] Active tracks: #14 [apple]: CONFIRMED | #15 [banana]: CONFIRMED
## [535132] Active tracks: #14 [apple]: CONFIRMED | #15 [banana]: CONFIRMED
## [535259] Active tracks: #14 [apple]: TEMPORARILY_MISSING | #15 [banana]: CONFIRMED
## [535384] Active tracks: #14 [apple]: TEMPORARILY_MISSING | #15 [banana]: CONFIRMED
## [535508] Active tracks: #14 [apple]: TEMPORARILY_MISSING | #15 [banana]: CONFIRMED
## [535633] Active tracks: #14 [apple]: TEMPORARILY_MISSING | #15 [banana]: CONFIRMED
## [535759] Active tracks: #15 [banana]: CONFIRMED
## [535883] Active tracks: #15 [banana]: CONFIRMED
## [536009] Active tracks: #15 [banana]: CONFIRMED
## [536133] Active tracks: #15 [banana]: CONFIRMED
## [536259] Active tracks: #15 [banana]: TEMPORARILY_MISSING | #16 [apple]: TENTATIVE
## [536383] Active tracks: #15 [banana]: TEMPORARILY_MISSING | #16 [apple]: TEMPORARILY_MISSING
## [536508] Active tracks: #15 [banana]: TEMPORARILY_MISSING | #16 [apple]: TEMPORARILY_MISSING
## [536633] Active tracks: #15 [banana]: TEMPORARILY_MISSING | #16 [apple]: TEMPORARILY_MISSING
## [536759] Active tracks: #15 [banana]: CONFIRMED | #16 [apple]: TENTATIVE
## [536883] Active tracks: #15 [banana]: TEMPORARILY_MISSING | #16 [apple]: CONFIRMED
## [537009] Active tracks: #15 [banana]: TEMPORARILY_MISSING | #16 [apple]: CONFIRMED
## [537134] Active tracks: #15 [banana]: TEMPORARILY_MISSING | #16 [apple]: TEMPORARILY_MISSING
## [537259] Active tracks: #15 [banana]: TEMPORARILY_MISSING | #16 [apple]: TEMPORARILY_MISSING
## [537384] Active tracks: #16 [apple]: TEMPORARILY_MISSING
## [537509] Active tracks: #16 [apple]: TEMPORARILY_MISSING
## [537634] Active tracks: none
## [537759] Active tracks: none
## [537884] Active tracks: none
## [538009] Active tracks: #17 [apple]: TENTATIVE
## [538134] Active tracks: #17 [apple]: CONFIRMED
## [538259] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [538384] Active tracks: #17 [apple]: CONFIRMED
## [538509] Active tracks: #17 [apple]: CONFIRMED
## [538634] Active tracks: #17 [apple]: CONFIRMED
## [538759] Active tracks: #17 [apple]: CONFIRMED
## [538885] Active tracks: #17 [apple]: CONFIRMED
## [539010] Active tracks: #17 [apple]: CONFIRMED
## [539134] Active tracks: #17 [apple]: CONFIRMED
## [539261] Active tracks: #17 [apple]: CONFIRMED
## [539385] Active tracks: #17 [apple]: CONFIRMED
## [539510] Active tracks: #17 [apple]: CONFIRMED
## [539635] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [539760] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [539885] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [540010] Active tracks: #17 [apple]: TEMPORARILY_MISSING
```

```
## [540136] Active tracks: #17 [apple]: CONFIRMED
## [540261] Active tracks: #17 [apple]: CONFIRMED
## [540385] Active tracks: #17 [apple]: TEMPORARILY_MISSING | #18 [banana]: TENTATIVE
## [540510] Active tracks: #17 [apple]: TEMPORARILY_MISSING | #18 [banana]: TEMPORARILY_MISSING
## [540635] Active tracks: #17 [apple]: CONFIRMED | #18 [banana]: TEMPORARILY_MISSING
## [540760] Active tracks: #17 [apple]: CONFIRMED | #18 [banana]: TEMPORARILY_MISSING
## [540886] Active tracks: #17 [apple]: TEMPORARILY_MISSING | #18 [banana]: TEMPORARILY_MISSING
## [541011] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [541136] Active tracks: #17 [apple]: CONFIRMED
## [541261] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [541386] Active tracks: #17 [apple]: CONFIRMED
## [541511] Active tracks: #17 [apple]: CONFIRMED
## [541637] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [541761] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [541886] Active tracks: #17 [apple]: CONFIRMED
## [542011] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [542137] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [542262] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [542387] Active tracks: #17 [apple]: TEMPORARILY_MISSING
## [542512] Active tracks: none
## [542637] Active tracks: none
## [542762] Active tracks: none
## [542888] Active tracks: none
## [543011] Active tracks: none
## [543137] Active tracks: none
## [543262] Active tracks: none
## [543387] Active tracks: none
## [543513] Active tracks: #19 [apple]: TENTATIVE
## [543638] Active tracks: #19 [apple]: CONFIRMED
## [543763] Active tracks: #19 [apple]: CONFIRMED
## [543888] Active tracks: #19 [apple]: TEMPORARILY_MISSING
## [544013] Active tracks: #19 [apple]: CONFIRMED
## [544138] Active tracks: #19 [apple]: CONFIRMED
## [544262] Active tracks: #19 [apple]: CONFIRMED
## [544388] Active tracks: #19 [apple]: CONFIRMED
## [544513] Active tracks: #19 [apple]: CONFIRMED
## [544638] Active tracks: #19 [apple]: CONFIRMED
## [544765] Active tracks: #19 [apple]: CONFIRMED
## [544889] Active tracks: #19 [apple]: CONFIRMED
## [545013] Active tracks: #19 [apple]: CONFIRMED
## [545140] Active tracks: #19 [apple]: TEMPORARILY_MISSING
## [545265] Active tracks: #19 [apple]: TEMPORARILY_MISSING
## [545389] Active tracks: #19 [apple]: CONFIRMED
## [545514] Active tracks: #19 [apple]: TEMPORARILY_MISSING
## [545639] Active tracks: #19 [apple]: CONFIRMED
## [545764] Active tracks: #19 [apple]: CONFIRMED
## [545889] Active tracks: #19 [apple]: CONFIRMED
## [546014] Active tracks: #19 [apple]: TEMPORARILY_MISSING
## [546139] Active tracks: #19 [apple]: TEMPORARILY_MISSING
## [546264] Active tracks: #19 [apple]: TEMPORARILY_MISSING
## [546389] Active tracks: #19 [apple]: TEMPORARILY_MISSING
## [546514] Active tracks: none
## [546639] Active tracks: #20 [apple]: TENTATIVE
## [546765] Active tracks: #20 [apple]: TEMPORARILY_MISSING
```

```
## [546890] Active tracks: #20 [apple]: TEMPORARILY_MISSING
## [547014] Active tracks: #20 [apple]: TEMPORARILY_MISSING
## [547140] Active tracks: #20 [apple]: TEMPORARILY_MISSING
## [547265] Active tracks: none
## [547390] Active tracks: #21 [apple]: TENTATIVE
## [547516] Active tracks: #21 [apple]: CONFIRMED
## [547641] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [547767] Active tracks: #21 [apple]: CONFIRMED
## [547891] Active tracks: #21 [apple]: CONFIRMED
## [548016] Active tracks: #21 [apple]: CONFIRMED
## [548141] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [548265] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [548391] Active tracks: #21 [apple]: CONFIRMED
## [548516] Active tracks: #21 [apple]: CONFIRMED
## [548641] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [548766] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [548891] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [549016] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [549141] Active tracks: #21 [apple]: CONFIRMED
## [549267] Active tracks: #21 [apple]: CONFIRMED
## [549391] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [549518] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [549641] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [549766] Active tracks: #21 [apple]: CONFIRMED
## [549892] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [550017] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [550142] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [550267] Active tracks: #21 [apple]: TEMPORARILY_MISSING
## [550392] Active tracks: none
## [550517] Active tracks: #22 [apple]: TENTATIVE
## [550643] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [550767] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [550894] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [551017] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [551142] Active tracks: #22 [apple]: TENTATIVE
## [551268] Active tracks: #22 [apple]: CONFIRMED
## [551393] Active tracks: #22 [apple]: CONFIRMED
## [551518] Active tracks: #22 [apple]: CONFIRMED
## [551643] Active tracks: #22 [apple]: CONFIRMED
## [551768] Active tracks: #22 [apple]: CONFIRMED
## [551893] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [552019] Active tracks: #22 [apple]: CONFIRMED
## [552144] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [552270] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [552393] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [552518] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [552644] Active tracks: #22 [apple]: CONFIRMED
## [552769] Active tracks: #22 [apple]: CONFIRMED
## [552894] Active tracks: #22 [apple]: CONFIRMED
## [553018] Active tracks: #22 [apple]: CONFIRMED
## [553143] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [553269] Active tracks: #22 [apple]: CONFIRMED
## [553394] Active tracks: #22 [apple]: CONFIRMED
## [553519] Active tracks: #22 [apple]: CONFIRMED
```

```
## [553644] Active tracks: #22 [apple]: CONFIRMED
## [553770] Active tracks: #22 [apple]: CONFIRMED
## [553894] Active tracks: #22 [apple]: CONFIRMED
## [554019] Active tracks: #22 [apple]: CONFIRMED
## [554144] Active tracks: #22 [apple]: CONFIRMED
## [554270] Active tracks: #22 [apple]: CONFIRMED
## [554395] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [554521] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [554645] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [554769] Active tracks: #22 [apple]: CONFIRMED
## [554895] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [555020] Active tracks: #22 [apple]: TEMPORARILY_MISSING
## [555145] Active tracks: #22 [apple]: TEMPORARILY_MISSING | #23 [bowl]: TENTATIVE
## [555270] Active tracks: #22 [apple]: TEMPORARILY_MISSING | #23 [bowl]: TEMPORARILY_MISSING
## [555395] Active tracks: #23 [bowl]: TEMPORARILY_MISSING
## [555520] Active tracks: #23 [bowl]: TEMPORARILY_MISSING | #24 [apple]: TENTATIVE
## [555645] Active tracks: #23 [bowl]: TEMPORARILY_MISSING | #24 [apple]: CONFIRMED
## [555771] Active tracks: #24 [apple]: CONFIRMED
## [555895] Active tracks: #24 [apple]: CONFIRMED
## [556021] Active tracks: #24 [apple]: CONFIRMED
## [556145] Active tracks: #24 [apple]: CONFIRMED
## [556271] Active tracks: #24 [apple]: CONFIRMED
## [556396] Active tracks: #24 [apple]: CONFIRMED
## [556521] Active tracks: #24 [apple]: CONFIRMED
## [556646] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [556771] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [556895] Active tracks: #24 [apple]: CONFIRMED
## [557021] Active tracks: #24 [apple]: CONFIRMED
## [557146] Active tracks: #24 [apple]: CONFIRMED
## [557271] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [557396] Active tracks: #24 [apple]: CONFIRMED
## [557522] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [557649] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [557772] Active tracks: #24 [apple]: CONFIRMED
## [557897] Active tracks: #24 [apple]: CONFIRMED
## [558022] Active tracks: #24 [apple]: CONFIRMED
## [558147] Active tracks: #24 [apple]: CONFIRMED
## [558271] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [558397] Active tracks: #24 [apple]: CONFIRMED
## [558522] Active tracks: #24 [apple]: CONFIRMED
## [558647] Active tracks: #24 [apple]: CONFIRMED
## [558773] Active tracks: #24 [apple]: CONFIRMED
## [558897] Active tracks: #24 [apple]: CONFIRMED
## [559022] Active tracks: #24 [apple]: CONFIRMED
## [559147] Active tracks: #24 [apple]: CONFIRMED
## [559272] Active tracks: #24 [apple]: CONFIRMED
## [559398] Active tracks: #24 [apple]: CONFIRMED
## [559522] Active tracks: #24 [apple]: CONFIRMED
## [559647] Active tracks: #24 [apple]: CONFIRMED
## [559772] Active tracks: #24 [apple]: CONFIRMED
## [559897] Active tracks: #24 [apple]: CONFIRMED
## [560022] Active tracks: #24 [apple]: CONFIRMED
## [560147] Active tracks: #24 [apple]: CONFIRMED
## [560273] Active tracks: #24 [apple]: CONFIRMED
```

```
## [560398] Active tracks: #24 [apple]: CONFIRMED
## [560523] Active tracks: #24 [apple]: CONFIRMED
## [560648] Active tracks: #24 [apple]: CONFIRMED
## [560773] Active tracks: #24 [apple]: CONFIRMED
## [560898] Active tracks: #24 [apple]: CONFIRMED
## [561023] Active tracks: #24 [apple]: CONFIRMED
## [561148] Active tracks: #24 [apple]: CONFIRMED
## [561273] Active tracks: #24 [apple]: CONFIRMED
## [561398] Active tracks: #24 [apple]: CONFIRMED
## [561523] Active tracks: #24 [apple]: CONFIRMED
## [561649] Active tracks: #24 [apple]: CONFIRMED
## [561774] Active tracks: #24 [apple]: CONFIRMED
## [561900] Active tracks: #24 [apple]: CONFIRMED
## [562025] Active tracks: #24 [apple]: CONFIRMED
## [562149] Active tracks: #24 [apple]: CONFIRMED
## [562276] Active tracks: #24 [apple]: CONFIRMED
## [562401] Active tracks: #24 [apple]: CONFIRMED
## [562525] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [562650] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [562775] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [562901] Active tracks: #24 [apple]: CONFIRMED
## [563024] Active tracks: #24 [apple]: CONFIRMED
## [563150] Active tracks: #24 [apple]: CONFIRMED
## [563275] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [563401] Active tracks: #24 [apple]: CONFIRMED
## [563525] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [563650] Active tracks: #24 [apple]: CONFIRMED
## [563774] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [563900] Active tracks: #24 [apple]: CONFIRMED
## [564026] Active tracks: #24 [apple]: CONFIRMED
## [564152] Active tracks: #24 [apple]: CONFIRMED
## [564277] Active tracks: #24 [apple]: CONFIRMED
## [564401] Active tracks: #24 [apple]: CONFIRMED
## [564526] Active tracks: #24 [apple]: CONFIRMED
## [564651] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [564776] Active tracks: #24 [apple]: CONFIRMED
## [564902] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [565026] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [565152] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [565276] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [565402] Active tracks: none
## [565527] Active tracks: #25 [apple]: TENTATIVE
## [565652] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [565777] Active tracks: #25 [apple]: TENTATIVE
## [565902] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [566027] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [566152] Active tracks: #25 [apple]: TENTATIVE
## [566277] Active tracks: #25 [apple]: CONFIRMED
## [566402] Active tracks: #25 [apple]: CONFIRMED
## [566527] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [566652] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [566778] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [566903] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [567028] Active tracks: none
```

```
## [567153] Active tracks: #26 [apple]: TENTATIVE
## [567278] Active tracks: #26 [apple]: TEMPORARILY_MISSING
## [567402] Active tracks: #26 [apple]: TEMPORARILY_MISSING
## [567528] Active tracks: #26 [apple]: TEMPORARILY_MISSING
## [567653] Active tracks: #26 [apple]: TEMPORARILY_MISSING
## [567778] Active tracks: none
## [567903] Active tracks: none
## [568028] Active tracks: none
## [568154] Active tracks: none
## [568279] Active tracks: none
## [568405] Active tracks: #27 [apple]: TENTATIVE
## [568529] Active tracks: #27 [apple]: TEMPORARILY_MISSING
## [568654] Active tracks: #27 [apple]: TEMPORARILY_MISSING
## [568779] Active tracks: #27 [apple]: TEMPORARILY_MISSING
## [568904] Active tracks: #27 [apple]: TEMPORARILY_MISSING
## [569029] Active tracks: none
## [569154] Active tracks: none
## [569279] Active tracks: none
## [569404] Active tracks: none
## [569530] Active tracks: none
## [569656] Active tracks: none
## [569780] Active tracks: none
## [569905] Active tracks: none
## [570029] Active tracks: none
## [570154] Active tracks: #28 [knife]: TENTATIVE
## [570279] Active tracks: #28 [knife]: TEMPORARILY_MISSING
## [570406] Active tracks: #28 [knife]: TEMPORARILY_MISSING
## [570530] Active tracks: #28 [knife]: TEMPORARILY_MISSING
## [570655] Active tracks: #28 [knife]: TEMPORARILY_MISSING
## [570780] Active tracks: none
## [570905] Active tracks: none
## [571031] Active tracks: #29 [apple]: TENTATIVE
## [571155] Active tracks: #29 [apple]: TEMPORARILY_MISSING
## [571281] Active tracks: #29 [apple]: TEMPORARILY_MISSING
## [571406] Active tracks: #29 [apple]: TEMPORARILY_MISSING
## [571530] Active tracks: #29 [apple]: TENTATIVE
## [571655] Active tracks: #29 [apple]: CONFIRMED
## [571780] Active tracks: #29 [apple]: CONFIRMED
## [571910] Active tracks: #29 [apple]: CONFIRMED
## [572035] Active tracks: #29 [apple]: CONFIRMED
## [572160] Active tracks: #29 [apple]: CONFIRMED
## [572285] Active tracks: #29 [apple]: CONFIRMED
## [572410] Active tracks: #29 [apple]: CONFIRMED
## [572536] Active tracks: #29 [apple]: CONFIRMED
## [572661] Active tracks: #29 [apple]: CONFIRMED
## [572786] Active tracks: #29 [apple]: CONFIRMED | #30 [banana]: TENTATIVE
## [572911] Active tracks: #29 [apple]: TEMPORARILY_MISSING | #30 [banana]: TEMPORARILY_MISSING
## [573035] Active tracks: #29 [apple]: TEMPORARILY_MISSING | #30 [banana]: TEMPORARILY_MISSING
## [573160] Active tracks: #29 [apple]: TEMPORARILY_MISSING | #30 [banana]: TEMPORARILY_MISSING
## [573285] Active tracks: #29 [apple]: TEMPORARILY_MISSING | #30 [banana]: TEMPORARILY_MISSING
## [573410] Active tracks: none
## [573535] Active tracks: none
## [573661] Active tracks: none
## [573787] Active tracks: none
```

```
## [573911] Active tracks: none
## [574037] Active tracks: none
## [574161] Active tracks: none
## [574286] Active tracks: none
## [574410] Active tracks: none
```

On measurement set 3 Experiment metadata:

```
cat(jsonlite::prettify(readLines("tracking_3_single_object_slowly_moving.jsonl", warn = FALSE, encoding
```

```
## {
##     "record_type": "metadata",
##     "schema_version": 1,
##     "experiment_id": "tracking_20261013T154144Z_357d54d9",
##     "scenario": "One trackable object is moved slowly in front of the camera",
##     "custom_notes": "",
##     "tracker_config": {
##         "CONFIRMATION_FRAMES": 2,
##         "MISSING_FRAMES_LIMIT": 5,
##         "LOOSE_MATCH_DISTANCE_PX": 40,
##         "TIGHT_MATCH_DISTANCE_PX": 15,
##         "BACKUP_MATCH_DISTANCE_PX": 60
##     },
##     "target_sampling_fps": 8,
##     "start_time_utc": "2026-10-13T15:41:44.530Z",
##     "end_time_utc": "2026-10-13T15:42:48.669Z",
##     "frame_count": 512
## }
##
```

Track timeline:

```
print_track_timeline(t3)
```

```
## [425307] Active tracks: #1 [apple]: TENTATIVE | #2 [knife]: TENTATIVE
## [425414] Active tracks: #1 [apple]: CONFIRMED | #2 [knife]: TEMPORARILY_MISSING
## [425537] Active tracks: #1 [apple]: CONFIRMED | #2 [knife]: TEMPORARILY_MISSING
## [425662] Active tracks: #1 [apple]: CONFIRMED | #2 [knife]: TEMPORARILY_MISSING
## [425787] Active tracks: #1 [apple]: CONFIRMED | #2 [knife]: TENTATIVE
## [425912] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #2 [knife]: TEMPORARILY_MISSING
## [426035] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #2 [knife]: TEMPORARILY_MISSING
## [426163] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #2 [knife]: TEMPORARILY_MISSING
## [426287] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #2 [knife]: TEMPORARILY_MISSING
## [426412] Active tracks: none
## [426537] Active tracks: none
## [426663] Active tracks: #3 [apple]: TENTATIVE
## [426787] Active tracks: #3 [apple]: TEMPORARILY_MISSING
## [426914] Active tracks: #3 [apple]: TENTATIVE
## [427038] Active tracks: #3 [apple]: CONFIRMED
## [427163] Active tracks: #3 [apple]: TEMPORARILY_MISSING
## [427288] Active tracks: #3 [apple]: CONFIRMED | #4 [knife]: TENTATIVE
## [427414] Active tracks: #3 [apple]: CONFIRMED | #4 [knife]: CONFIRMED
## [427538] Active tracks: #3 [apple]: CONFIRMED | #4 [knife]: TEMPORARILY_MISSING
## [427663] Active tracks: #3 [apple]: CONFIRMED | #4 [knife]: TEMPORARILY_MISSING
## [427788] Active tracks: #3 [apple]: CONFIRMED | #4 [knife]: TEMPORARILY_MISSING
## [427914] Active tracks: #3 [apple]: CONFIRMED | #4 [knife]: TEMPORARILY_MISSING
```

```
## [428039] Active tracks: #3 [apple]: CONFIRMED
## [428165] Active tracks: #3 [apple]: CONFIRMED
## [428288] Active tracks: #3 [apple]: CONFIRMED
## [428415] Active tracks: #3 [apple]: CONFIRMED
## [428538] Active tracks: #3 [apple]: CONFIRMED
## [428663] Active tracks: #3 [apple]: CONFIRMED
## [428788] Active tracks: #3 [apple]: CONFIRMED
## [428915] Active tracks: #3 [apple]: CONFIRMED
## [429039] Active tracks: #3 [apple]: CONFIRMED
## [429165] Active tracks: #3 [apple]: CONFIRMED
## [429290] Active tracks: #3 [apple]: CONFIRMED
## [429416] Active tracks: #3 [apple]: CONFIRMED
## [429539] Active tracks: #3 [apple]: CONFIRMED
## [429666] Active tracks: #3 [apple]: CONFIRMED
## [429789] Active tracks: #3 [apple]: CONFIRMED
## [429915] Active tracks: #3 [apple]: CONFIRMED
## [430040] Active tracks: #3 [apple]: CONFIRMED | #5 [knife]: TENTATIVE
## [430165] Active tracks: #3 [apple]: CONFIRMED | #5 [knife]: TEMPORARILY_MISSING
## [430289] Active tracks: #3 [apple]: CONFIRMED | #5 [knife]: TEMPORARILY_MISSING
## [430415] Active tracks: #3 [apple]: CONFIRMED | #5 [knife]: TEMPORARILY_MISSING
## [430540] Active tracks: #3 [apple]: CONFIRMED | #5 [knife]: TEMPORARILY_MISSING
## [430667] Active tracks: #3 [apple]: CONFIRMED
## [430789] Active tracks: #3 [apple]: CONFIRMED
## [430916] Active tracks: #3 [apple]: TEMPORARILY_MISSING
## [431040] Active tracks: #3 [apple]: CONFIRMED
## [431165] Active tracks: #3 [apple]: TEMPORARILY_MISSING
## [431291] Active tracks: #3 [apple]: TEMPORARILY_MISSING
## [431417] Active tracks: #3 [apple]: TEMPORARILY_MISSING
## [431541] Active tracks: #3 [apple]: TEMPORARILY_MISSING
## [431667] Active tracks: none
## [431791] Active tracks: none
## [431916] Active tracks: none
## [432041] Active tracks: none
## [432167] Active tracks: none
## [432290] Active tracks: none
## [432416] Active tracks: none
## [432541] Active tracks: none
## [432666] Active tracks: none
## [432790] Active tracks: #6 [apple]: TENTATIVE
## [432916] Active tracks: #6 [apple]: CONFIRMED
## [433042] Active tracks: #6 [apple]: CONFIRMED
## [433167] Active tracks: #6 [apple]: CONFIRMED
## [433290] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [433416] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [433541] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [433667] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [433791] Active tracks: none
## [433917] Active tracks: none
## [434042] Active tracks: none
## [434168] Active tracks: none
## [434292] Active tracks: none
## [434417] Active tracks: none
## [434543] Active tracks: none
## [434667] Active tracks: none
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## [434918] Active tracks: none
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## [436168] Active tracks: none
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## [436542] Active tracks: none
## [436668] Active tracks: none
## [436793] Active tracks: none
## [436918] Active tracks: none
## [437043] Active tracks: none
## [437168] Active tracks: none
## [437293] Active tracks: none
## [437418] Active tracks: none
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## [437919] Active tracks: none
## [438045] Active tracks: none
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## [438795] Active tracks: none
## [438921] Active tracks: none
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## [439170] Active tracks: none
## [439295] Active tracks: none
## [439422] Active tracks: none
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## [439671] Active tracks: none
## [439794] Active tracks: none
## [439920] Active tracks: none
## [440045] Active tracks: none
## [440170] Active tracks: none
## [440295] Active tracks: none
## [440420] Active tracks: none
## [440544] Active tracks: none
## [440670] Active tracks: none
## [440795] Active tracks: none
## [440921] Active tracks: none
## [441047] Active tracks: #7 [apple]: TENTATIVE
## [441171] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [441296] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [441421] Active tracks: #7 [apple]: TEMPORARILY_MISSING
```

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## [441547] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [441672] Active tracks: none
## [441797] Active tracks: none
## [441923] Active tracks: #8 [apple]: TENTATIVE
## [442047] Active tracks: #8 [apple]: CONFIRMED
## [442171] Active tracks: #8 [apple]: CONFIRMED
## [442296] Active tracks: #8 [apple]: CONFIRMED
## [442423] Active tracks: #8 [apple]: TEMPORARILY_MISSING
## [442548] Active tracks: #8 [apple]: TEMPORARILY_MISSING
## [442673] Active tracks: #8 [apple]: TEMPORARILY_MISSING
## [442798] Active tracks: #8 [apple]: TEMPORARILY_MISSING
## [442923] Active tracks: #8 [apple]: CONFIRMED
## [443047] Active tracks: #8 [apple]: TEMPORARILY_MISSING
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## [443298] Active tracks: #8 [apple]: TEMPORARILY_MISSING
## [443423] Active tracks: #8 [apple]: TEMPORARILY_MISSING
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## [443798] Active tracks: none
## [443925] Active tracks: none
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## [445675] Active tracks: #9 [apple]: CONFIRMED
## [445799] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [445925] Active tracks: #9 [apple]: CONFIRMED
## [446050] Active tracks: #9 [apple]: CONFIRMED
## [446175] Active tracks: #9 [apple]: CONFIRMED
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## [446424] Active tracks: #9 [apple]: CONFIRMED
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## [446924] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [447049] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [447175] Active tracks: #9 [apple]: CONFIRMED
## [447299] Active tracks: #9 [apple]: CONFIRMED
## [447424] Active tracks: #9 [apple]: CONFIRMED
## [447550] Active tracks: #9 [apple]: CONFIRMED
## [447675] Active tracks: #9 [apple]: CONFIRMED
## [447800] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [447925] Active tracks: #9 [apple]: CONFIRMED
## [448050] Active tracks: #9 [apple]: CONFIRMED
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## [448300] Active tracks: #9 [apple]: TEMPORARILY_MISSING
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## [448677] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [448801] Active tracks: #9 [apple]: CONFIRMED
## [448925] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [449053] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [449176] Active tracks: #9 [apple]: CONFIRMED
## [449301] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [449427] Active tracks: #9 [apple]: CONFIRMED
## [449550] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [449677] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [449805] Active tracks: #9 [apple]: TEMPORARILY_MISSING
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## [451427] Active tracks: #10 [apple]: CONFIRMED
## [451554] Active tracks: #10 [apple]: TEMPORARILY_MISSING
## [451677] Active tracks: #10 [apple]: TEMPORARILY_MISSING
## [451802] Active tracks: #10 [apple]: TEMPORARILY_MISSING
## [451928] Active tracks: #10 [apple]: CONFIRMED
## [452053] Active tracks: #10 [apple]: TEMPORARILY_MISSING
## [452178] Active tracks: #10 [apple]: TEMPORARILY_MISSING
## [452303] Active tracks: #10 [apple]: CONFIRMED
## [452428] Active tracks: #10 [apple]: CONFIRMED
## [452553] Active tracks: #10 [apple]: CONFIRMED
## [452678] Active tracks: #10 [apple]: CONFIRMED
## [452803] Active tracks: #10 [apple]: CONFIRMED
## [452928] Active tracks: #10 [apple]: CONFIRMED
## [453053] Active tracks: #10 [apple]: CONFIRMED
## [453180] Active tracks: #10 [apple]: TEMPORARILY_MISSING
## [453304] Active tracks: #10 [apple]: TEMPORARILY_MISSING
## [453428] Active tracks: #10 [apple]: TEMPORARILY_MISSING
## [453553] Active tracks: #10 [apple]: TEMPORARILY_MISSING
## [453678] Active tracks: none
## [453805] Active tracks: none
## [453929] Active tracks: none
## [454054] Active tracks: none
## [454179] Active tracks: none
## [454304] Active tracks: none
## [454429] Active tracks: none
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## [454679] Active tracks: none
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## [454929] Active tracks: none
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## [455555] Active tracks: #11 [apple]: CONFIRMED
## [455680] Active tracks: #11 [apple]: TEMPORARILY_MISSING
## [455805] Active tracks: #11 [apple]: TEMPORARILY_MISSING
## [455930] Active tracks: #11 [apple]: TEMPORARILY_MISSING
## [456055] Active tracks: #11 [apple]: TEMPORARILY_MISSING
## [456179] Active tracks: none
## [456305] Active tracks: none
## [456430] Active tracks: none
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## [456679] Active tracks: none
## [456805] Active tracks: none
## [456930] Active tracks: none
## [457055] Active tracks: none
## [457180] Active tracks: none
## [457305] Active tracks: none
## [457431] Active tracks: none
## [457555] Active tracks: none
## [457680] Active tracks: none
## [457806] Active tracks: none
## [457932] Active tracks: none
## [458057] Active tracks: none
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## [458431] Active tracks: none
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## [458681] Active tracks: none
## [458806] Active tracks: none
## [458931] Active tracks: none
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## [459557] Active tracks: #12 [apple]: CONFIRMED
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## [459807] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [459932] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [460057] Active tracks: #12 [apple]: CONFIRMED
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## [460934] Active tracks: #12 [apple]: CONFIRMED
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## [461184] Active tracks: #12 [apple]: CONFIRMED
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## [461435] Active tracks: #12 [apple]: CONFIRMED
## [461560] Active tracks: #12 [apple]: CONFIRMED
## [461683] Active tracks: #12 [apple]: CONFIRMED
```

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## [461809] Active tracks: #12 [apple]: CONFIRMED
## [461933] Active tracks: #12 [apple]: CONFIRMED
## [462058] Active tracks: #12 [apple]: CONFIRMED
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## [462309] Active tracks: #12 [apple]: CONFIRMED
## [462435] Active tracks: #12 [apple]: CONFIRMED
## [462560] Active tracks: #12 [apple]: CONFIRMED
## [462684] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [462810] Active tracks: #12 [apple]: CONFIRMED
## [462935] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [463060] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [463185] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [463309] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [463435] Active tracks: #12 [apple]: CONFIRMED
## [463561] Active tracks: #12 [apple]: CONFIRMED
## [463684] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [463810] Active tracks: #12 [apple]: CONFIRMED
## [463935] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [464061] Active tracks: #12 [apple]: CONFIRMED
## [464185] Active tracks: #12 [apple]: CONFIRMED
## [464310] Active tracks: #12 [apple]: CONFIRMED
## [464435] Active tracks: #12 [apple]: CONFIRMED
## [464561] Active tracks: #12 [apple]: CONFIRMED
## [464685] Active tracks: #12 [apple]: CONFIRMED
## [464811] Active tracks: #12 [apple]: CONFIRMED
## [464935] Active tracks: #12 [apple]: CONFIRMED
## [465061] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [465186] Active tracks: #12 [apple]: CONFIRMED
## [465311] Active tracks: #12 [apple]: CONFIRMED
## [465436] Active tracks: #12 [apple]: CONFIRMED
## [465561] Active tracks: #12 [apple]: CONFIRMED
## [465686] Active tracks: #12 [apple]: CONFIRMED
## [465812] Active tracks: #12 [apple]: CONFIRMED
## [465936] Active tracks: #12 [apple]: CONFIRMED
## [466061] Active tracks: #12 [apple]: CONFIRMED
## [466187] Active tracks: #12 [apple]: CONFIRMED
## [466311] Active tracks: #12 [apple]: CONFIRMED
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## [466812] Active tracks: #12 [apple]: CONFIRMED
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## [467062] Active tracks: #12 [apple]: CONFIRMED
## [467187] Active tracks: #12 [apple]: CONFIRMED
## [467311] Active tracks: #12 [apple]: CONFIRMED
## [467437] Active tracks: #12 [apple]: CONFIRMED
## [467562] Active tracks: #12 [apple]: CONFIRMED
## [467687] Active tracks: #12 [apple]: CONFIRMED
## [467812] Active tracks: #12 [apple]: CONFIRMED
## [467937] Active tracks: #12 [apple]: CONFIRMED
## [468063] Active tracks: #12 [apple]: CONFIRMED
## [468187] Active tracks: #12 [apple]: CONFIRMED
## [468312] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [468437] Active tracks: #12 [apple]: TEMPORARILY_MISSING
```

```
## [468563] Active tracks: #12 [apple]: TEMPORARILY_MISSING
## [468688] Active tracks: #12 [apple]: TEMPORARILY_MISSING
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## [468938] Active tracks: none
## [469063] Active tracks: none
## [469188] Active tracks: none
## [469312] Active tracks: none
## [469438] Active tracks: none
## [469563] Active tracks: none
## [469688] Active tracks: none
## [469813] Active tracks: none
## [469938] Active tracks: none
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## [470188] Active tracks: none
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## [470440] Active tracks: #13 [apple]: CONFIRMED
## [470565] Active tracks: #13 [apple]: CONFIRMED
## [470689] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [470814] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [470940] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [471065] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [471190] Active tracks: #13 [apple]: CONFIRMED
## [471315] Active tracks: #13 [apple]: CONFIRMED
## [471440] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [471565] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [471690] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [471815] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [471940] Active tracks: #13 [apple]: CONFIRMED
## [472065] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [472190] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [472315] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [472440] Active tracks: #13 [apple]: TEMPORARILY_MISSING
## [472566] Active tracks: none
## [472692] Active tracks: none
## [472816] Active tracks: none
## [472941] Active tracks: none
## [473066] Active tracks: #14 [apple]: TENTATIVE
## [473192] Active tracks: #14 [apple]: TEMPORARILY_MISSING
## [473316] Active tracks: #14 [apple]: TENTATIVE
## [473440] Active tracks: #14 [apple]: CONFIRMED
## [473566] Active tracks: #14 [apple]: TEMPORARILY_MISSING
## [473690] Active tracks: #14 [apple]: TEMPORARILY_MISSING
## [473816] Active tracks: #14 [apple]: TEMPORARILY_MISSING
## [473941] Active tracks: #14 [apple]: TEMPORARILY_MISSING
## [474068] Active tracks: none
## [474193] Active tracks: none
## [474317] Active tracks: none
## [474441] Active tracks: none
## [474567] Active tracks: none
## [474693] Active tracks: none
## [474818] Active tracks: none
## [474943] Active tracks: none
## [475068] Active tracks: none
## [475193] Active tracks: #15 [apple]: TENTATIVE
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```
## [475317] Active tracks: #15 [apple]: CONFIRMED
## [475442] Active tracks: #15 [apple]: CONFIRMED
## [475567] Active tracks: #15 [apple]: CONFIRMED
## [475692] Active tracks: #15 [apple]: CONFIRMED
## [475817] Active tracks: #15 [apple]: CONFIRMED
## [475942] Active tracks: #15 [apple]: CONFIRMED
## [476068] Active tracks: #15 [apple]: CONFIRMED
## [476193] Active tracks: #15 [apple]: CONFIRMED
## [476318] Active tracks: #15 [apple]: CONFIRMED
## [476443] Active tracks: #15 [apple]: CONFIRMED
## [476568] Active tracks: #15 [apple]: CONFIRMED
## [476694] Active tracks: #15 [apple]: CONFIRMED
## [476818] Active tracks: #15 [apple]: CONFIRMED
## [476944] Active tracks: #15 [apple]: CONFIRMED
## [477069] Active tracks: #15 [apple]: CONFIRMED
## [477194] Active tracks: #15 [apple]: CONFIRMED
## [477319] Active tracks: #15 [apple]: CONFIRMED
## [477443] Active tracks: #15 [apple]: CONFIRMED
## [477569] Active tracks: #15 [apple]: CONFIRMED
## [477695] Active tracks: #15 [apple]: CONFIRMED
## [477820] Active tracks: #15 [apple]: CONFIRMED
## [477944] Active tracks: #15 [apple]: CONFIRMED
## [478069] Active tracks: #15 [apple]: CONFIRMED
## [478195] Active tracks: #15 [apple]: CONFIRMED
## [478319] Active tracks: #15 [apple]: CONFIRMED
## [478444] Active tracks: #15 [apple]: CONFIRMED
## [478569] Active tracks: #15 [apple]: CONFIRMED
## [478694] Active tracks: #15 [apple]: CONFIRMED
## [478820] Active tracks: #15 [apple]: CONFIRMED
## [478944] Active tracks: #15 [apple]: CONFIRMED
## [479070] Active tracks: #15 [apple]: CONFIRMED
## [479195] Active tracks: #15 [apple]: CONFIRMED
## [479321] Active tracks: #15 [apple]: CONFIRMED
## [479446] Active tracks: #15 [apple]: CONFIRMED
## [479571] Active tracks: #15 [apple]: CONFIRMED
## [479696] Active tracks: #15 [apple]: CONFIRMED
## [479821] Active tracks: #15 [apple]: CONFIRMED
## [479946] Active tracks: #15 [apple]: CONFIRMED
## [480072] Active tracks: #15 [apple]: CONFIRMED
## [480197] Active tracks: #15 [apple]: CONFIRMED
## [480322] Active tracks: #15 [apple]: CONFIRMED
## [480447] Active tracks: #15 [apple]: CONFIRMED
## [480572] Active tracks: #15 [apple]: CONFIRMED
## [480697] Active tracks: #15 [apple]: CONFIRMED
## [480822] Active tracks: #15 [apple]: CONFIRMED
## [480947] Active tracks: #15 [apple]: CONFIRMED
## [481072] Active tracks: #15 [apple]: CONFIRMED
## [481197] Active tracks: #15 [apple]: CONFIRMED
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## [481455] Active tracks: #15 [apple]: CONFIRMED
## [481579] Active tracks: #15 [apple]: CONFIRMED
## [481704] Active tracks: #15 [apple]: CONFIRMED
## [481829] Active tracks: #15 [apple]: CONFIRMED
## [481955] Active tracks: #15 [apple]: CONFIRMED
```

```
## [482080] Active tracks: #15 [apple]: CONFIRMED
## [482206] Active tracks: #15 [apple]: CONFIRMED
## [482328] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [482455] Active tracks: #15 [apple]: CONFIRMED
## [482580] Active tracks: #15 [apple]: CONFIRMED
## [482705] Active tracks: #15 [apple]: CONFIRMED
## [482830] Active tracks: #15 [apple]: CONFIRMED
## [482954] Active tracks: #15 [apple]: CONFIRMED
## [483080] Active tracks: #15 [apple]: CONFIRMED
## [483206] Active tracks: #15 [apple]: CONFIRMED
## [483330] Active tracks: #15 [apple]: CONFIRMED
## [483454] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [483580] Active tracks: #15 [apple]: CONFIRMED
## [483704] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [483830] Active tracks: #15 [apple]: CONFIRMED
## [483954] Active tracks: #15 [apple]: CONFIRMED
## [484080] Active tracks: #15 [apple]: CONFIRMED
## [484205] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [484331] Active tracks: #15 [apple]: CONFIRMED
## [484455] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [484580] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [484705] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [484830] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [484955] Active tracks: none
## [485081] Active tracks: none
## [485205] Active tracks: none
## [485331] Active tracks: none
## [485457] Active tracks: none
## [485581] Active tracks: #16 [apple]: TENTATIVE
## [485706] Active tracks: #16 [apple]: CONFIRMED
## [485831] Active tracks: #16 [apple]: CONFIRMED
## [485955] Active tracks: #16 [apple]: TEMPORARILY_MISSING
## [486081] Active tracks: #16 [apple]: CONFIRMED
## [486206] Active tracks: #16 [apple]: CONFIRMED
## [486332] Active tracks: #16 [apple]: CONFIRMED
## [486456] Active tracks: #16 [apple]: CONFIRMED
## [486582] Active tracks: #16 [apple]: CONFIRMED
## [486707] Active tracks: #16 [apple]: TEMPORARILY_MISSING
## [486832] Active tracks: #16 [apple]: TEMPORARILY_MISSING
## [486957] Active tracks: #16 [apple]: CONFIRMED
## [487083] Active tracks: #16 [apple]: CONFIRMED
## [487208] Active tracks: #16 [apple]: CONFIRMED
## [487333] Active tracks: #16 [apple]: CONFIRMED
## [487457] Active tracks: #16 [apple]: CONFIRMED
## [487582] Active tracks: #16 [orange (fruit)]: CONFIRMED
## [487708] Active tracks: #16 [orange (fruit)]: CONFIRMED
## [487831] Active tracks: #16 [orange (fruit)]: TEMPORARILY_MISSING
## [487958] Active tracks: #16 [orange (fruit)]: TEMPORARILY_MISSING
## [488082] Active tracks: #16 [orange (fruit)]: TEMPORARILY_MISSING
## [488208] Active tracks: #16 [orange (fruit)]: TEMPORARILY_MISSING
## [488332] Active tracks: none
## [488458] Active tracks: none
## [488583] Active tracks: none
## [488708] Active tracks: none
```

```
## [488832] Active tracks: none
## [488959] Active tracks: none
## [489083] Active tracks: none
## [489209] Active tracks: none
```

On measurement set 4 Experiment metadata:

```
cat(jsonlite::prettify(readLines("tracking_4_single_object_rapidly_moving.jsonl", warn = FALSE, encodin
```

```
## {
##     "record_type": "metadata",
##     "schema_version": 1,
##     "experiment_id": "tracking_20261013T155401Z_fd8b76d4",
##     "scenario": "One trackable object is moved rapidly in front of the camera",
##     "custom_notes": "",
##     "tracker_config": {
##         "CONFIRMATION_FRAMES": 2,
##         "MISSING_FRAMES_LIMIT": 5,
##         "LOOSE_MATCH_DISTANCE_PX": 40,
##         "TIGHT_MATCH_DISTANCE_PX": 15,
##         "BACKUP_MATCH_DISTANCE_PX": 60
##     },
##     "target_sampling_fps": 8,
##     "start_time_utc": "2026-10-13T15:54:01.131Z",
##     "end_time_utc": "2026-10-13T15:55:03.264Z",
##     "frame_count": 496
## }
##
```

Track timeline:

```
print_track_timeline(t4)
```

```
## [7212] Active tracks: none
## [7346] Active tracks: none
## [7424] Active tracks: none
## [7550] Active tracks: none
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## [9300] Active tracks: #2 [knife]: TEMPORARILY_MISSING | #3 [bowl]: CONFIRMED
## [9425] Active tracks: #3 [bowl]: CONFIRMED
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## [9925] Active tracks: #3 [bowl]: TEMPORARILY_MISSING | #4 [knife]: TENTATIVE
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## [12803] Active tracks: #6 [bowl]: TEMPORARILY_MISSING
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## [13179] Active tracks: #6 [bowl]: TEMPORARILY_MISSING | #7 [spoon]: TEMPORARILY_MISSING
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## [20809] Active tracks: #12 [bowl]: CONFIRMED
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## [21058] Active tracks: #12 [bowl]: TEMPORARILY_MISSING
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## [21433] Active tracks: #12 [bowl]: TEMPORARILY_MISSING | #13 [knife]: TENTATIVE
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## [22684] Active tracks: #13 [knife]: TEMPORARILY_MISSING | #14 [bowl]: CONFIRMED
## [22810] Active tracks: #14 [bowl]: CONFIRMED
## [22934] Active tracks: #14 [bowl]: CONFIRMED
## [23060] Active tracks: #14 [bowl]: CONFIRMED
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## [26936] Active tracks: #15 [bowl]: TEMPORARILY_MISSING
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## [27312] Active tracks: #16 [bowl]: CONFIRMED
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## [28562] Active tracks: #16 [bowl]: TEMPORARILY_MISSING | #17 [knife]: TEMPORARILY_MISSING
## [28687] Active tracks: #16 [bowl]: TEMPORARILY_MISSING | #17 [knife]: TEMPORARILY_MISSING
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## [32567] Active tracks: #21 [bowl]: CONFIRMED
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## [33567] Active tracks: #22 [bowl]: CONFIRMED
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## [35693] Active tracks: #23 [bowl]: TEMPORARILY_MISSING
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## [35946] Active tracks: #23 [bowl]: TEMPORARILY_MISSING | #24 [bowl]: TEMPORARILY_MISSING
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## [36569] Active tracks: #25 [bowl]: TEMPORARILY_MISSING
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## [39696] Active tracks: #27 [bowl]: CONFIRMED
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## [40070] Active tracks: #27 [bowl]: TEMPORARILY_MISSING
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## [40696] Active tracks: #28 [bowl]: CONFIRMED
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## [42948] Active tracks: #29 [knife]: TEMPORARILY_MISSING
## [43072] Active tracks: #29 [knife]: TENTATIVE
## [43197] Active tracks: #29 [knife]: TEMPORARILY_MISSING
## [43324] Active tracks: #29 [knife]: TEMPORARILY_MISSING
## [43448] Active tracks: #29 [knife]: TEMPORARILY_MISSING
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## [44075] Active tracks: #30 [bowl]: TEMPORARILY_MISSING | #31 [spoon]: TEMPORARILY_MISSING
## [44199] Active tracks: #31 [spoon]: TEMPORARILY_MISSING
## [44323] Active tracks: #31 [spoon]: TEMPORARILY_MISSING
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## [46451] Active tracks: #32 [bowl]: TEMPORARILY_MISSING | #33 [bowl]: TENTATIVE
## [46576] Active tracks: #32 [bowl]: TEMPORARILY_MISSING | #33 [bowl]: CONFIRMED
## [46700] Active tracks: #33 [bowl]: TEMPORARILY_MISSING
## [46826] Active tracks: #33 [bowl]: TEMPORARILY_MISSING
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## [48451] Active tracks: #34 [bowl]: TEMPORARILY_MISSING
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## [49953] Active tracks: #35 [bowl]: TEMPORARILY_MISSING
## [50076] Active tracks: #35 [bowl]: TEMPORARILY_MISSING | #36 [bowl]: TENTATIVE
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## [50327] Active tracks: #36 [bowl]: CONFIRMED
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## [50702] Active tracks: #36 [bowl]: CONFIRMED
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## [53705] Active tracks: #38 [bowl]: TEMPORARILY_MISSING
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## [57582] Active tracks: #41 [bowl]: TENTATIVE
## [57709] Active tracks: #41 [bowl]: TEMPORARILY_MISSING
## [57834] Active tracks: #41 [bowl]: TEMPORARILY_MISSING
## [57958] Active tracks: #41 [bowl]: TEMPORARILY_MISSING | #42 [bowl]: TENTATIVE
## [58083] Active tracks: #41 [bowl]: TEMPORARILY_MISSING | #42 [bowl]: CONFIRMED
## [58208] Active tracks: #42 [bowl]: TEMPORARILY_MISSING
## [58334] Active tracks: #42 [bowl]: TEMPORARILY_MISSING
## [58459] Active tracks: #42 [bowl]: TEMPORARILY_MISSING
## [58583] Active tracks: #42 [bowl]: TEMPORARILY_MISSING
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## [59335] Active tracks: #43 [bowl]: CONFIRMED
## [59459] Active tracks: #43 [bowl]: TEMPORARILY_MISSING
## [59584] Active tracks: #43 [bowl]: TEMPORARILY_MISSING
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## [60335] Active tracks: #44 [bowl]: TEMPORARILY_MISSING
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## [60585] Active tracks: #44 [bowl]: TEMPORARILY_MISSING
## [60710] Active tracks: #45 [bowl]: TENTATIVE
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## [60960] Active tracks: #45 [bowl]: TEMPORARILY_MISSING
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## [62587] Active tracks: #46 [bowl]: CONFIRMED
## [62712] Active tracks: #46 [bowl]: TEMPORARILY_MISSING
## [62837] Active tracks: #46 [bowl]: TEMPORARILY_MISSING | #47 [bowl]: TENTATIVE
## [62961] Active tracks: #46 [bowl]: TEMPORARILY_MISSING | #47 [bowl]: CONFIRMED
## [63087] Active tracks: #46 [bowl]: TEMPORARILY_MISSING | #47 [bowl]: CONFIRMED
## [63212] Active tracks: #47 [bowl]: CONFIRMED
## [63337] Active tracks: #47 [bowl]: CONFIRMED
## [63462] Active tracks: #47 [bowl]: CONFIRMED
## [63587] Active tracks: #47 [bowl]: CONFIRMED
## [63712] Active tracks: #47 [bowl]: TEMPORARILY_MISSING
## [63837] Active tracks: #47 [bowl]: TEMPORARILY_MISSING
## [63962] Active tracks: #47 [bowl]: TEMPORARILY_MISSING
```

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## [64462] Active tracks: none
## [64588] Active tracks: none
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## [64963] Active tracks: #48 [bowl]: TEMPORARILY_MISSING
## [65087] Active tracks: #48 [bowl]: TEMPORARILY_MISSING
## [65213] Active tracks: #48 [bowl]: TEMPORARILY_MISSING
## [65338] Active tracks: #48 [bowl]: TEMPORARILY_MISSING | #49 [bowl]: TENTATIVE
## [65463] Active tracks: #49 [bowl]: TEMPORARILY_MISSING
## [65589] Active tracks: #49 [bowl]: TEMPORARILY_MISSING
## [65714] Active tracks: #49 [bowl]: TEMPORARILY_MISSING | #50 [bowl]: TENTATIVE
## [65838] Active tracks: #49 [bowl]: TEMPORARILY_MISSING | #50 [bowl]: TEMPORARILY_MISSING
## [65964] Active tracks: #50 [bowl]: TEMPORARILY_MISSING
## [66089] Active tracks: #50 [bowl]: TEMPORARILY_MISSING | #51 [bowl]: TENTATIVE
## [66214] Active tracks: #50 [bowl]: TEMPORARILY_MISSING | #51 [bowl]: TEMPORARILY_MISSING
## [66339] Active tracks: #51 [bowl]: TEMPORARILY_MISSING
## [66464] Active tracks: #51 [bowl]: TEMPORARILY_MISSING
## [66589] Active tracks: #51 [bowl]: TEMPORARILY_MISSING | #52 [bowl]: TENTATIVE
## [66715] Active tracks: #52 [bowl]: CONFIRMED
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## [69091] Active tracks: none
```

On measurement set 5 Experiment metadata:

```
cat(jsonlite::prettify(readLines("tracking_5_single_object_enter_frame.jsonl", warn = FALSE, encoding =
```

```
## {
##     "record_type": "metadata",
##     "schema_version": 1,
##     "experiment_id": "tracking_20261013T155944Z_f42aae8d",
##     "scenario": "One trackable object enters the frame at some point during this experiment",
##     "custom_notes": "",
##     "tracker_config": {
##         "CONFIRMATION_FRAMES": 2,
##         "MISSING_FRAMES_LIMIT": 5,
```

```
##         "LOOSE_MATCH_DISTANCE_PX": 40,
##         "TIGHT_MATCH_DISTANCE_PX": 15,
##         "BACKUP_MATCH_DISTANCE_PX": 60
##     },
##     "target_sampling_fps": 8,
##     "start_time_utc": "2026-10-13T15:59:44.775Z",
##     "end_time_utc": "2026-10-13T16:00:48.282Z",
##     "frame_count": 507
## }
##
```

Track timeline:

```
print_track_timeline(t5)
```

```
## [8234] Active tracks: none
## [8296] Active tracks: none
## [8419] Active tracks: none
## [8544] Active tracks: none
## [8669] Active tracks: none
## [8794] Active tracks: none
## [8918] Active tracks: none
## [9045] Active tracks: none
## [9170] Active tracks: none
## [9296] Active tracks: none
## [9419] Active tracks: none
## [9545] Active tracks: none
## [9669] Active tracks: none
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## [12545] Active tracks: none
## [12671] Active tracks: none
## [12796] Active tracks: none
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## [13047] Active tracks: none
```

```
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## [15049] Active tracks: none
## [15172] Active tracks: none
## [15297] Active tracks: none
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## [63701] Active tracks: #2 [bowl]: CONFIRMED
## [63826] Active tracks: #2 [bowl]: CONFIRMED
## [63951] Active tracks: #2 [bowl]: CONFIRMED
## [64077] Active tracks: #2 [bowl]: CONFIRMED
## [64201] Active tracks: #2 [bowl]: CONFIRMED
## [64328] Active tracks: #2 [bowl]: CONFIRMED
## [64451] Active tracks: #2 [bowl]: CONFIRMED
## [64576] Active tracks: #2 [bowl]: CONFIRMED
## [64701] Active tracks: #2 [bowl]: CONFIRMED
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## [64951] Active tracks: #2 [bowl]: CONFIRMED
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## [65201] Active tracks: #2 [bowl]: CONFIRMED
## [65327] Active tracks: #2 [bowl]: CONFIRMED
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## [66077] Active tracks: #2 [bowl]: CONFIRMED
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## [66327] Active tracks: #2 [bowl]: CONFIRMED
## [66452] Active tracks: #2 [bowl]: CONFIRMED
## [66577] Active tracks: #2 [bowl]: CONFIRMED
## [66703] Active tracks: #2 [bowl]: CONFIRMED
## [66828] Active tracks: #2 [bowl]: CONFIRMED
## [66953] Active tracks: #2 [bowl]: CONFIRMED
## [67079] Active tracks: #2 [bowl]: CONFIRMED
```

```
## [67203] Active tracks: #2 [bowl]: CONFIRMED
## [67327] Active tracks: #2 [bowl]: CONFIRMED
## [67453] Active tracks: #2 [bowl]: CONFIRMED
## [67578] Active tracks: #2 [bowl]: CONFIRMED
## [67703] Active tracks: #2 [bowl]: CONFIRMED
## [67828] Active tracks: #2 [bowl]: CONFIRMED
## [67954] Active tracks: #2 [bowl]: CONFIRMED
## [68079] Active tracks: #2 [bowl]: CONFIRMED
## [68204] Active tracks: #2 [bowl]: CONFIRMED
## [68328] Active tracks: #2 [bowl]: CONFIRMED
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## [68954] Active tracks: #2 [bowl]: CONFIRMED
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## [69204] Active tracks: #2 [bowl]: CONFIRMED
## [69329] Active tracks: #2 [bowl]: CONFIRMED
## [69454] Active tracks: #2 [bowl]: CONFIRMED
## [69579] Active tracks: #2 [bowl]: TEMPORARILY_MISSING
## [69704] Active tracks: #2 [bowl]: CONFIRMED
## [69828] Active tracks: #2 [bowl]: CONFIRMED
## [69954] Active tracks: #2 [bowl]: CONFIRMED
## [70081] Active tracks: #2 [bowl]: CONFIRMED
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## [70329] Active tracks: #2 [bowl]: CONFIRMED
## [70455] Active tracks: #2 [bowl]: CONFIRMED
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## [70705] Active tracks: #2 [bowl]: CONFIRMED
## [70829] Active tracks: #2 [bowl]: CONFIRMED
## [70954] Active tracks: #2 [bowl]: CONFIRMED
## [71080] Active tracks: #2 [bowl]: CONFIRMED
## [71204] Active tracks: #2 [bowl]: CONFIRMED
## [71330] Active tracks: #2 [bowl]: CONFIRMED
## [71455] Active tracks: #2 [bowl]: CONFIRMED
```

On measurement set 6 Experiment metadata:

```
cat(jsonlite::prettify(readLines("tracking_6_single_object_leave_frame.jsonl", warn = FALSE, encoding =
```


```
## {
##     "record_type": "metadata",
##     "schema_version": 1,
##     "experiment_id": "tracking_20261013T160446Z_e9c9b18b",
##     "scenario": "One trackable object leaves the frame at some point during this experiment",
##     "custom_notes": "",
##     "tracker_config": {
##         "CONFIRMATION_FRAMES": 2,
##         "MISSING_FRAMES_LIMIT": 5,
##         "LOOSE_MATCH_DISTANCE_PX": 40,
##         "TIGHT_MATCH_DISTANCE_PX": 15,
##         "BACKUP_MATCH_DISTANCE_PX": 60
##     },
##     "target_sampling_fps": 8,
##     "start_time_utc": "2026-10-13T16:04:46.292Z",
```

```
##     "end_time_utc": "2026-10-13T16:05:46.786Z",
##     "frame_count": 483
## }
##
```

Track timeline:

```
print_track_timeline(t6)
```

```
## [231793] Active tracks: none
## [231848] Active tracks: none
## [231971] Active tracks: none
## [232096] Active tracks: none
## [232221] Active tracks: none
## [232356] Active tracks: none
## [232482] Active tracks: #1 [knife]: TENTATIVE
## [232606] Active tracks: #1 [knife]: TEMPORARILY_MISSING
## [232730] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #2 [knife]: TENTATIVE
## [232855] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #2 [knife]: TEMPORARILY_MISSING
## [232981] Active tracks: #1 [knife]: TENTATIVE | #2 [knife]: TEMPORARILY_MISSING
## [233106] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #2 [knife]: TEMPORARILY_MISSING
## [233231] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #2 [knife]: TEMPORARILY_MISSING | #3 [knif
## [233356] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #3 [knife]: CONFIRMED
## [233481] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #3 [knife]: TEMPORARILY_MISSING
## [233606] Active tracks: #1 [knife]: TENTATIVE | #3 [knife]: TEMPORARILY_MISSING
## [233731] Active tracks: #1 [knife]: CONFIRMED | #3 [knife]: TEMPORARILY_MISSING
## [233856] Active tracks: #1 [knife]: CONFIRMED | #3 [knife]: TEMPORARILY_MISSING
## [233982] Active tracks: #1 [knife]: CONFIRMED
## [234107] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #4 [knife]: TENTATIVE
## [234232] Active tracks: #1 [knife]: CONFIRMED | #4 [knife]: TEMPORARILY_MISSING
## [234356] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #4 [knife]: TENTATIVE
## [234481] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #4 [knife]: TEMPORARILY_MISSING
## [234607] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #4 [knife]: TEMPORARILY_MISSING
## [234732] Active tracks: #1 [knife]: TEMPORARILY_MISSING | #4 [knife]: TEMPORARILY_MISSING
## [234857] Active tracks: #4 [knife]: TEMPORARILY_MISSING
## [234982] Active tracks: none
## [235107] Active tracks: none
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## [235358] Active tracks: #5 [knife]: TEMPORARILY_MISSING
## [235481] Active tracks: #5 [knife]: TENTATIVE
## [235606] Active tracks: #5 [knife]: TEMPORARILY_MISSING | #6 [knife]: TENTATIVE
## [235731] Active tracks: #5 [knife]: TEMPORARILY_MISSING | #6 [knife]: CONFIRMED
## [235856] Active tracks: #5 [knife]: TEMPORARILY_MISSING | #6 [knife]: CONFIRMED
## [235982] Active tracks: #5 [knife]: TEMPORARILY_MISSING | #6 [knife]: TEMPORARILY_MISSING
## [236106] Active tracks: #6 [knife]: TEMPORARILY_MISSING
## [236232] Active tracks: #6 [knife]: TEMPORARILY_MISSING
## [236357] Active tracks: #6 [knife]: TEMPORARILY_MISSING
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## [236607] Active tracks: #7 [knife]: TEMPORARILY_MISSING
## [236731] Active tracks: #7 [knife]: TEMPORARILY_MISSING
## [236857] Active tracks: #7 [knife]: TEMPORARILY_MISSING | #8 [knife]: TENTATIVE
## [236983] Active tracks: #7 [knife]: TEMPORARILY_MISSING | #8 [knife]: TEMPORARILY_MISSING
## [237107] Active tracks: #8 [knife]: TEMPORARILY_MISSING
## [237233] Active tracks: #8 [knife]: TENTATIVE
## [237359] Active tracks: #8 [knife]: CONFIRMED
```

```
## [237483] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TENTATIVE
## [237609] Active tracks: #8 [knife]: TEMPORARILY_MISSING | #9 [knife]: TEMPORARILY_MISSING
## [237733] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TEMPORARILY_MISSING
## [237858] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TENTATIVE
## [237983] Active tracks: #8 [knife]: TEMPORARILY_MISSING | #9 [knife]: TEMPORARILY_MISSING
## [238109] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TEMPORARILY_MISSING
## [238235] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TEMPORARILY_MISSING
## [238360] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TEMPORARILY_MISSING
## [238485] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TENTATIVE
## [238610] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TEMPORARILY_MISSING
## [238734] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TEMPORARILY_MISSING
## [238860] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TEMPORARILY_MISSING
## [238986] Active tracks: #8 [knife]: CONFIRMED | #9 [knife]: TEMPORARILY_MISSING
## [239111] Active tracks: #8 [knife]: CONFIRMED
## [239235] Active tracks: #8 [knife]: CONFIRMED
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## [239485] Active tracks: #8 [knife]: CONFIRMED
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## [239861] Active tracks: #8 [knife]: CONFIRMED
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## [240360] Active tracks: #8 [knife]: TEMPORARILY_MISSING
## [240486] Active tracks: #8 [knife]: TEMPORARILY_MISSING
## [240611] Active tracks: #8 [knife]: TEMPORARILY_MISSING
## [240736] Active tracks: #8 [knife]: TEMPORARILY_MISSING
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## [240985] Active tracks: none
## [241112] Active tracks: none
## [241236] Active tracks: none
## [241361] Active tracks: none
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## [241611] Active tracks: #10 [knife]: TEMPORARILY_MISSING
## [241736] Active tracks: #10 [knife]: TEMPORARILY_MISSING
## [241861] Active tracks: #10 [knife]: TEMPORARILY_MISSING
## [241987] Active tracks: #10 [knife]: TEMPORARILY_MISSING
## [242113] Active tracks: none
## [242237] Active tracks: none
## [242363] Active tracks: #11 [knife]: TENTATIVE
## [242487] Active tracks: #11 [knife]: TEMPORARILY_MISSING | #12 [knife]: TENTATIVE
## [242612] Active tracks: #11 [knife]: TEMPORARILY_MISSING | #12 [knife]: CONFIRMED
## [242737] Active tracks: #11 [knife]: TEMPORARILY_MISSING | #12 [knife]: TEMPORARILY_MISSING | #13 [kn
## [242862] Active tracks: #11 [knife]: TEMPORARILY_MISSING | #12 [knife]: TEMPORARILY_MISSING | #13 [kn
## [242988] Active tracks: #12 [knife]: TEMPORARILY_MISSING | #13 [knife]: TEMPORARILY_MISSING | #14 [kn
## [243114] Active tracks: #12 [knife]: TEMPORARILY_MISSING | #13 [knife]: TEMPORARILY_MISSING | #14 [kn
## [243237] Active tracks: #13 [knife]: TEMPORARILY_MISSING | #14 [knife]: TEMPORARILY_MISSING | #15 [kn
## [243364] Active tracks: #13 [knife]: TEMPORARILY_MISSING | #14 [knife]: TEMPORARILY_MISSING | #15 [kn
## [243487] Active tracks: #14 [knife]: TEMPORARILY_MISSING | #15 [knife]: TEMPORARILY_MISSING | #16 [kn
## [243612] Active tracks: #15 [knife]: TEMPORARILY_MISSING | #16 [knife]: TEMPORARILY_MISSING
## [243738] Active tracks: #15 [knife]: TEMPORARILY_MISSING | #16 [knife]: TENTATIVE
## [243864] Active tracks: #16 [knife]: TEMPORARILY_MISSING | #17 [knife]: TENTATIVE
## [243988] Active tracks: #16 [knife]: TEMPORARILY_MISSING | #17 [knife]: CONFIRMED
## [244114] Active tracks: #16 [knife]: TEMPORARILY_MISSING | #17 [knife]: TEMPORARILY_MISSING
```

```
## [244238] Active tracks: #16 [knife]: TEMPORARILY_MISSING | #17 [knife]: TEMPORARILY_MISSING | #18 [kn
## [244363] Active tracks: #17 [knife]: TEMPORARILY_MISSING | #18 [knife]: TEMPORARILY_MISSING
## [244488] Active tracks: #17 [knife]: TEMPORARILY_MISSING | #18 [knife]: TEMPORARILY_MISSING
## [244613] Active tracks: #18 [knife]: TEMPORARILY_MISSING
## [244739] Active tracks: #18 [knife]: TEMPORARILY_MISSING | #19 [knife]: TENTATIVE
## [244864] Active tracks: #19 [knife]: TEMPORARILY_MISSING
## [244989] Active tracks: #19 [knife]: TEMPORARILY_MISSING
## [245113] Active tracks: #19 [knife]: TEMPORARILY_MISSING
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## [245489] Active tracks: none
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## [245990] Active tracks: none
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## [246740] Active tracks: none
## [246865] Active tracks: none
## [246990] Active tracks: none
## [247116] Active tracks: none
## [247241] Active tracks: none
## [247365] Active tracks: none
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## [247615] Active tracks: none
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## [247865] Active tracks: none
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## [248491] Active tracks: none
## [248616] Active tracks: none
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## [249367] Active tracks: #20 [knife]: TEMPORARILY_MISSING
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## [249742] Active tracks: #20 [knife]: TEMPORARILY_MISSING
## [249867] Active tracks: #20 [knife]: TENTATIVE
## [249992] Active tracks: #20 [knife]: TEMPORARILY_MISSING
## [250116] Active tracks: #20 [knife]: TEMPORARILY_MISSING
## [250242] Active tracks: #20 [knife]: TEMPORARILY_MISSING | #21 [knife]: TENTATIVE
## [250367] Active tracks: #20 [knife]: TEMPORARILY_MISSING | #21 [knife]: CONFIRMED
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## [250617] Active tracks: #21 [knife]: TEMPORARILY_MISSING | #22 [knife]: TEMPORARILY_MISSING
## [250742] Active tracks: #21 [knife]: TEMPORARILY_MISSING | #22 [knife]: TEMPORARILY_MISSING
## [250867] Active tracks: #21 [knife]: TEMPORARILY_MISSING | #22 [knife]: TEMPORARILY_MISSING
```

```
## [250993] Active tracks: #22 [knife]: TEMPORARILY_MISSING
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## [251243] Active tracks: #22 [knife]: TEMPORARILY_MISSING | #23 [knife]: TENTATIVE
## [251367] Active tracks: #22 [knife]: TENTATIVE | #23 [knife]: TEMPORARILY_MISSING
## [251500] Active tracks: #22 [knife]: CONFIRMED | #23 [knife]: TEMPORARILY_MISSING
## [251625] Active tracks: #22 [knife]: TEMPORARILY_MISSING | #23 [knife]: TEMPORARILY_MISSING
## [251750] Active tracks: #22 [knife]: CONFIRMED | #23 [knife]: TEMPORARILY_MISSING
## [251875] Active tracks: #22 [knife]: CONFIRMED
## [252001] Active tracks: #22 [knife]: TEMPORARILY_MISSING
## [252126] Active tracks: #22 [knife]: CONFIRMED
## [252250] Active tracks: #22 [knife]: CONFIRMED
## [252375] Active tracks: #22 [knife]: CONFIRMED
## [252500] Active tracks: #22 [knife]: CONFIRMED
## [252625] Active tracks: #22 [knife]: CONFIRMED
## [252751] Active tracks: #22 [knife]: TEMPORARILY_MISSING
## [252876] Active tracks: #22 [knife]: CONFIRMED
## [253001] Active tracks: #22 [knife]: CONFIRMED
## [253126] Active tracks: #22 [knife]: TEMPORARILY_MISSING
## [253251] Active tracks: #22 [knife]: CONFIRMED
## [253376] Active tracks: #22 [knife]: CONFIRMED
## [253501] Active tracks: #22 [knife]: CONFIRMED
## [253625] Active tracks: #22 [knife]: CONFIRMED
## [253751] Active tracks: #22 [knife]: CONFIRMED
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## [254001] Active tracks: #22 [knife]: CONFIRMED
## [254127] Active tracks: #22 [knife]: CONFIRMED
## [254252] Active tracks: #22 [knife]: CONFIRMED
## [254377] Active tracks: #22 [knife]: CONFIRMED
## [254501] Active tracks: #22 [knife]: TEMPORARILY_MISSING
## [254626] Active tracks: #22 [knife]: CONFIRMED
## [254752] Active tracks: #22 [knife]: CONFIRMED
## [254877] Active tracks: #22 [knife]: CONFIRMED
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## [255127] Active tracks: #22 [knife]: CONFIRMED
## [255253] Active tracks: #22 [knife]: CONFIRMED
## [255377] Active tracks: #22 [knife]: CONFIRMED
## [255502] Active tracks: #22 [knife]: TEMPORARILY_MISSING
## [255627] Active tracks: #22 [knife]: CONFIRMED
## [255753] Active tracks: #22 [knife]: CONFIRMED
## [255878] Active tracks: #22 [knife]: CONFIRMED
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## [256128] Active tracks: #22 [knife]: CONFIRMED
## [256253] Active tracks: #22 [knife]: TEMPORARILY_MISSING
## [256379] Active tracks: #22 [knife]: TEMPORARILY_MISSING
## [256503] Active tracks: #22 [knife]: TEMPORARILY_MISSING
## [256628] Active tracks: #22 [knife]: TEMPORARILY_MISSING
## [256755] Active tracks: none
## [256878] Active tracks: #24 [knife]: TENTATIVE
## [257003] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [257129] Active tracks: #24 [knife]: TENTATIVE
## [257254] Active tracks: #24 [knife]: CONFIRMED
## [257379] Active tracks: #24 [knife]: CONFIRMED
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## [257630] Active tracks: #24 [knife]: TEMPORARILY_MISSING
```

```
## [257755] Active tracks: #24 [knife]: CONFIRMED
## [257879] Active tracks: #24 [knife]: CONFIRMED
## [258004] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [258129] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [258254] Active tracks: #24 [knife]: CONFIRMED
## [258379] Active tracks: #24 [knife]: CONFIRMED
## [258504] Active tracks: #24 [knife]: CONFIRMED
## [258630] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [258755] Active tracks: #24 [knife]: CONFIRMED
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## [259130] Active tracks: #24 [knife]: CONFIRMED
## [259255] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [259380] Active tracks: #24 [knife]: CONFIRMED
## [259505] Active tracks: #24 [knife]: CONFIRMED
## [259632] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [259755] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [259881] Active tracks: #24 [knife]: CONFIRMED
## [260005] Active tracks: #24 [knife]: CONFIRMED
## [260130] Active tracks: #24 [knife]: CONFIRMED
## [260255] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [260380] Active tracks: #24 [knife]: CONFIRMED
## [260506] Active tracks: #24 [knife]: CONFIRMED
## [260631] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [260756] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [260881] Active tracks: #24 [knife]: CONFIRMED
## [261006] Active tracks: #24 [knife]: CONFIRMED
## [261131] Active tracks: #24 [knife]: CONFIRMED
## [261256] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [261381] Active tracks: #24 [knife]: CONFIRMED
## [261507] Active tracks: #24 [knife]: CONFIRMED
## [261631] Active tracks: #24 [knife]: CONFIRMED
## [261757] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [261881] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [262006] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [262131] Active tracks: #24 [knife]: CONFIRMED
## [262256] Active tracks: #24 [knife]: TEMPORARILY_MISSING
## [262381] Active tracks: #24 [knife]: TEMPORARILY_MISSING | #25 [spoon]: TENTATIVE
## [262506] Active tracks: #24 [knife]: TEMPORARILY_MISSING | #25 [spoon]: TEMPORARILY_MISSING
## [262632] Active tracks: #24 [knife]: TEMPORARILY_MISSING | #25 [spoon]: TENTATIVE
## [262757] Active tracks: #25 [spoon]: CONFIRMED
## [262882] Active tracks: #25 [spoon]: CONFIRMED
## [263009] Active tracks: #25 [spoon]: CONFIRMED
## [263132] Active tracks: #25 [spoon]: TEMPORARILY_MISSING
## [263258] Active tracks: #25 [spoon]: CONFIRMED
## [263382] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TENTATIVE
## [263508] Active tracks: #25 [spoon]: CONFIRMED | #26 [knife]: CONFIRMED
## [263633] Active tracks: #25 [spoon]: TEMPORARILY_MISSING | #26 [knife]: CONFIRMED
## [263758] Active tracks: #25 [spoon]: TEMPORARILY_MISSING | #26 [knife]: CONFIRMED
## [263883] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [264009] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [264132] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [264258] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [264383] Active tracks: #25 [spoon]: CONFIRMED | #26 [knife]: CONFIRMED
```

```
## [264508] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [264633] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [264758] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [264883] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [265008] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [265134] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [265259] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [265384] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [265508] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [265633] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [265759] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [265883] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [266010] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [266134] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [266259] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [266384] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [266510] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
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## [266760] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [266890] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [267009] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [267134] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [267259] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [267385] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [267510] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [267635] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [267760] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [267885] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [268012] Active tracks: #25 [knife]: TEMPORARILY_MISSING | #26 [knife]: CONFIRMED
## [268136] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: CONFIRMED
## [268260] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [268385] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [268511] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [268636] Active tracks: #25 [knife]: CONFIRMED | #26 [knife]: TEMPORARILY_MISSING
## [268761] Active tracks: #25 [knife]: CONFIRMED
## [268886] Active tracks: #25 [knife]: CONFIRMED
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## [269137] Active tracks: #25 [knife]: CONFIRMED
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## [269385] Active tracks: #25 [knife]: CONFIRMED
## [269511] Active tracks: #25 [knife]: CONFIRMED
## [269636] Active tracks: #25 [knife]: CONFIRMED
## [269761] Active tracks: #25 [knife]: CONFIRMED
## [269886] Active tracks: #25 [knife]: CONFIRMED
## [270011] Active tracks: #25 [knife]: CONFIRMED | #27 [knife]: TENTATIVE
## [270137] Active tracks: #25 [knife]: CONFIRMED | #27 [knife]: TEMPORARILY_MISSING
## [270262] Active tracks: #25 [knife]: CONFIRMED | #27 [knife]: TEMPORARILY_MISSING
## [270386] Active tracks: #25 [knife]: CONFIRMED | #27 [knife]: TEMPORARILY_MISSING
## [270512] Active tracks: #25 [knife]: CONFIRMED | #27 [knife]: TEMPORARILY_MISSING
## [270637] Active tracks: #25 [knife]: CONFIRMED
## [270762] Active tracks: #25 [knife]: TEMPORARILY_MISSING | #28 [knife]: TENTATIVE
## [270887] Active tracks: #25 [knife]: TEMPORARILY_MISSING | #28 [knife]: CONFIRMED
## [271012] Active tracks: #25 [knife]: TEMPORARILY_MISSING | #28 [knife]: CONFIRMED
## [271137] Active tracks: #25 [knife]: CONFIRMED | #28 [knife]: TEMPORARILY_MISSING
```

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## [271262] Active tracks: #25 [knife]: CONFIRMED | #28 [knife]: TEMPORARILY_MISSING
## [271387] Active tracks: #25 [knife]: CONFIRMED | #28 [knife]: TEMPORARILY_MISSING
## [271512] Active tracks: #25 [spoon]: CONFIRMED | #28 [knife]: TEMPORARILY_MISSING
## [271637] Active tracks: #25 [spoon]: CONFIRMED
## [271763] Active tracks: #25 [spoon]: CONFIRMED
## [271887] Active tracks: #25 [spoon]: CONFIRMED
## [272012] Active tracks: #25 [knife]: CONFIRMED
## [272137] Active tracks: #25 [knife]: CONFIRMED
## [272263] Active tracks: #25 [knife]: CONFIRMED
## [272388] Active tracks: #25 [knife]: CONFIRMED
## [272513] Active tracks: #25 [knife]: CONFIRMED
## [272638] Active tracks: #25 [knife]: CONFIRMED
## [272763] Active tracks: #25 [spoon]: CONFIRMED
## [272888] Active tracks: #25 [spoon]: CONFIRMED
## [273014] Active tracks: #25 [spoon]: TEMPORARILY_MISSING
## [273138] Active tracks: #25 [spoon]: CONFIRMED
## [273263] Active tracks: #25 [spoon]: CONFIRMED
## [273389] Active tracks: #25 [spoon]: TEMPORARILY_MISSING
## [273513] Active tracks: #25 [spoon]: CONFIRMED
## [273639] Active tracks: #25 [spoon]: TEMPORARILY_MISSING
## [273764] Active tracks: #25 [spoon]: TEMPORARILY_MISSING
## [273888] Active tracks: #25 [spoon]: TEMPORARILY_MISSING
## [274014] Active tracks: #25 [spoon]: TEMPORARILY_MISSING
## [274139] Active tracks: none
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## [274765] Active tracks: none
## [274889] Active tracks: none
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## [275140] Active tracks: none
## [275265] Active tracks: none
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## [275639] Active tracks: none
## [275765] Active tracks: none
## [275890] Active tracks: none
## [276015] Active tracks: none
## [276140] Active tracks: none
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## [276390] Active tracks: none
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## [276640] Active tracks: none
## [276766] Active tracks: none
## [276891] Active tracks: none
## [277016] Active tracks: none
## [277141] Active tracks: none
## [277266] Active tracks: none
## [277391] Active tracks: none
## [277517] Active tracks: none
## [277641] Active tracks: none
## [277766] Active tracks: none
## [277891] Active tracks: none
```

```
## [278017] Active tracks: none
## [278141] Active tracks: none
## [278266] Active tracks: none
## [278392] Active tracks: none
## [278517] Active tracks: none
## [278641] Active tracks: none
## [278767] Active tracks: none
## [278892] Active tracks: none
## [279017] Active tracks: none
## [279142] Active tracks: none
## [279267] Active tracks: none
## [279392] Active tracks: none
## [279517] Active tracks: none
## [279642] Active tracks: none
## [279767] Active tracks: none
## [279893] Active tracks: none
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## [280645] Active tracks: none
## [280768] Active tracks: none
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## [281019] Active tracks: none
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## [281269] Active tracks: none
## [281394] Active tracks: none
## [281519] Active tracks: none
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## [281895] Active tracks: none
## [282019] Active tracks: none
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## [282269] Active tracks: none
## [282395] Active tracks: none
## [282520] Active tracks: none
## [282645] Active tracks: none
## [282769] Active tracks: none
## [282895] Active tracks: none
## [283020] Active tracks: none
## [283145] Active tracks: none
## [283270] Active tracks: none
## [283396] Active tracks: none
## [283520] Active tracks: none
## [283645] Active tracks: none
## [283770] Active tracks: none
## [283896] Active tracks: none
## [284021] Active tracks: none
## [284146] Active tracks: none
## [284271] Active tracks: none
## [284395] Active tracks: none
## [284520] Active tracks: none
## [284646] Active tracks: none
```

```
## [284771] Active tracks: none
## [284896] Active tracks: none
## [285021] Active tracks: none
## [285146] Active tracks: none
## [285271] Active tracks: none
## [285396] Active tracks: none
## [285521] Active tracks: none
## [285646] Active tracks: none
## [285771] Active tracks: none
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## [286272] Active tracks: none
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## [287273] Active tracks: none
## [287398] Active tracks: none
## [287523] Active tracks: none
## [287647] Active tracks: none
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## [288150] Active tracks: none
## [288276] Active tracks: none
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## [288524] Active tracks: none
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## [288773] Active tracks: none
## [288898] Active tracks: none
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## [289148] Active tracks: none
## [289273] Active tracks: none
## [289398] Active tracks: none
## [289523] Active tracks: none
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## [290149] Active tracks: none
## [290275] Active tracks: none
## [290400] Active tracks: none
## [290525] Active tracks: none
## [290651] Active tracks: none
## [290774] Active tracks: none
## [290900] Active tracks: none
## [291024] Active tracks: none
## [291150] Active tracks: none
## [291275] Active tracks: none
## [291400] Active tracks: none
```

```
## [291525] Active tracks: none
## [291650] Active tracks: none
## [291775] Active tracks: none
## [291901] Active tracks: none
## [292027] Active tracks: none
```

On measurement set 7 Experiment metadata:

```
cat(jsonlite::prettify(readLines("tracking_7_multiple_objects_slowly_moving.jsonl", warn = FALSE, encodi
```


```
## {
##     "record_type": "metadata",
##     "schema_version": 1,
##     "experiment_id": "tracking_20261013T161018Z_162c0cd8",
##     "scenario": "Multiple trackable objects of the same tracking category are moved slowly in front
##     "custom_notes": "",
##     "tracker_config": {
##         "CONFIRMATION_FRAMES": 2,
##         "MISSING_FRAMES_LIMIT": 5,
##         "LOOSE_MATCH_DISTANCE_PX": 40,
##         "TIGHT_MATCH_DISTANCE_PX": 15,
##         "BACKUP_MATCH_DISTANCE_PX": 60
##     },
##     "target_sampling_fps": 8,
##     "start_time_utc": "2026-10-13T16:10:18.737Z",
##     "end_time_utc": "2026-10-13T16:11:21.824Z",
##     "frame_count": 504
## }
##
```


Track timeline:

```
print_track_timeline(t7)
```

```
## [8157] Active tracks: none
## [8300] Active tracks: none
## [8373] Active tracks: #1 [apple]: TENTATIVE
## [8498] Active tracks: #1 [apple]: CONFIRMED
## [8621] Active tracks: #1 [apple]: TEMPORARILY_MISSING
## [8746] Active tracks: #1 [apple]: CONFIRMED | #2 [apple]: TENTATIVE
## [8871] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #2 [apple]: TEMPORARILY_MISSING
## [8998] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #2 [apple]: TEMPORARILY_MISSING
## [9122] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #2 [apple]: TEMPORARILY_MISSING
## [9247] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #2 [apple]: TEMPORARILY_MISSING
## [9373] Active tracks: #1 [apple]: CONFIRMED
## [9498] Active tracks: #1 [apple]: CONFIRMED
## [9622] Active tracks: #1 [apple]: TEMPORARILY_MISSING
## [9748] Active tracks: #1 [apple]: CONFIRMED
## [9872] Active tracks: #1 [apple]: CONFIRMED
## [9998] Active tracks: #1 [apple]: CONFIRMED
## [10122] Active tracks: #1 [apple]: CONFIRMED | #3 [apple]: TENTATIVE
## [10247] Active tracks: #1 [apple]: CONFIRMED | #3 [apple]: CONFIRMED
## [10373] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #3 [apple]: CONFIRMED
## [10498] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #3 [apple]: TEMPORARILY_MISSING
## [10623] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #3 [apple]: TEMPORARILY_MISSING
## [10751] Active tracks: #1 [apple]: TEMPORARILY_MISSING | #3 [apple]: TEMPORARILY_MISSING
```

```
## [10873] Active tracks: #3 [apple]: TEMPORARILY_MISSING | #4 [apple]: TENTATIVE
## [10999] Active tracks: #4 [apple]: TEMPORARILY_MISSING | #5 [apple]: TENTATIVE
## [11124] Active tracks: #4 [apple]: TEMPORARILY_MISSING | #5 [apple]: TEMPORARILY_MISSING
## [11249] Active tracks: #4 [apple]: TEMPORARILY_MISSING | #5 [apple]: TENTATIVE
## [11374] Active tracks: #4 [apple]: TEMPORARILY_MISSING | #5 [apple]: CONFIRMED | #6 [apple]: TENTATI
## [11498] Active tracks: #5 [apple]: CONFIRMED | #6 [apple]: CONFIRMED
## [11623] Active tracks: #5 [apple]: CONFIRMED | #6 [apple]: CONFIRMED
## [11750] Active tracks: #5 [apple]: CONFIRMED | #6 [apple]: CONFIRMED
## [11873] Active tracks: #5 [apple]: CONFIRMED | #6 [apple]: CONFIRMED
## [12000] Active tracks: #5 [apple]: CONFIRMED | #6 [apple]: CONFIRMED
## [12125] Active tracks: #5 [apple]: TEMPORARILY_MISSING | #6 [apple]: CONFIRMED
## [12251] Active tracks: #5 [apple]: TEMPORARILY_MISSING | #6 [apple]: CONFIRMED
## [12375] Active tracks: #5 [apple]: TEMPORARILY_MISSING | #6 [apple]: CONFIRMED
## [12500] Active tracks: #5 [apple]: TEMPORARILY_MISSING | #6 [apple]: CONFIRMED
## [12624] Active tracks: #6 [apple]: CONFIRMED
## [12750] Active tracks: #6 [apple]: CONFIRMED
## [12874] Active tracks: #6 [apple]: CONFIRMED
## [13000] Active tracks: #6 [apple]: CONFIRMED
## [13124] Active tracks: #6 [apple]: CONFIRMED
## [13249] Active tracks: #6 [apple]: CONFIRMED
## [13375] Active tracks: #6 [apple]: CONFIRMED
## [13500] Active tracks: #6 [apple]: CONFIRMED
## [13624] Active tracks: #6 [apple]: CONFIRMED
## [13750] Active tracks: #6 [apple]: CONFIRMED
## [13874] Active tracks: #6 [apple]: CONFIRMED
## [14002] Active tracks: #6 [apple]: CONFIRMED
## [14124] Active tracks: #6 [apple]: CONFIRMED
## [14250] Active tracks: #6 [apple]: CONFIRMED
## [14375] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [14500] Active tracks: #6 [apple]: CONFIRMED
## [14625] Active tracks: #6 [apple]: CONFIRMED
## [14750] Active tracks: #6 [apple]: CONFIRMED
## [14876] Active tracks: #6 [apple]: CONFIRMED
## [15002] Active tracks: #6 [apple]: CONFIRMED
## [15125] Active tracks: #6 [apple]: CONFIRMED
## [15256] Active tracks: #6 [apple]: CONFIRMED
## [15376] Active tracks: #6 [apple]: CONFIRMED
## [15501] Active tracks: #6 [apple]: CONFIRMED
## [15626] Active tracks: #6 [apple]: CONFIRMED
## [15751] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [15876] Active tracks: #6 [apple]: CONFIRMED
## [16001] Active tracks: #6 [apple]: CONFIRMED
## [16126] Active tracks: #6 [apple]: CONFIRMED
## [16251] Active tracks: #6 [apple]: CONFIRMED
## [16377] Active tracks: #6 [apple]: CONFIRMED
## [16501] Active tracks: #6 [apple]: CONFIRMED
## [16627] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [16751] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [16877] Active tracks: #6 [apple]: CONFIRMED
## [17001] Active tracks: #6 [apple]: CONFIRMED
## [17127] Active tracks: #6 [apple]: CONFIRMED
## [17252] Active tracks: #6 [apple]: CONFIRMED
## [17377] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [17502] Active tracks: #6 [apple]: TEMPORARILY_MISSING
```

```
## [17627] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [17752] Active tracks: #6 [apple]: TEMPORARILY_MISSING
## [17877] Active tracks: none
## [18003] Active tracks: #7 [apple]: TENTATIVE
## [18127] Active tracks: #7 [apple]: CONFIRMED
## [18253] Active tracks: #7 [apple]: CONFIRMED
## [18378] Active tracks: #7 [apple]: CONFIRMED
## [18503] Active tracks: #7 [apple]: CONFIRMED
## [18628] Active tracks: #7 [apple]: CONFIRMED
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## [18879] Active tracks: #7 [apple]: CONFIRMED
## [19003] Active tracks: #7 [apple]: CONFIRMED
## [19128] Active tracks: #7 [apple]: CONFIRMED
## [19255] Active tracks: #7 [apple]: CONFIRMED
## [19379] Active tracks: #7 [apple]: CONFIRMED
## [19506] Active tracks: #7 [apple]: CONFIRMED
## [19630] Active tracks: #7 [apple]: CONFIRMED
## [19755] Active tracks: #7 [apple]: CONFIRMED
## [19879] Active tracks: #7 [apple]: CONFIRMED
## [20004] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [20130] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [20256] Active tracks: #7 [apple]: CONFIRMED | #8 [knife]: TENTATIVE
## [20381] Active tracks: #7 [apple]: TEMPORARILY_MISSING | #8 [knife]: TEMPORARILY_MISSING
## [20506] Active tracks: #7 [apple]: TEMPORARILY_MISSING | #8 [knife]: TENTATIVE
## [20630] Active tracks: #7 [apple]: TEMPORARILY_MISSING | #8 [knife]: CONFIRMED
## [20756] Active tracks: #7 [apple]: CONFIRMED | #8 [knife]: TEMPORARILY_MISSING
## [20880] Active tracks: #7 [apple]: TEMPORARILY_MISSING | #8 [knife]: TEMPORARILY_MISSING
## [21005] Active tracks: #7 [apple]: CONFIRMED | #8 [knife]: TEMPORARILY_MISSING
## [21130] Active tracks: #7 [apple]: TEMPORARILY_MISSING | #8 [knife]: TEMPORARILY_MISSING
## [21256] Active tracks: #7 [apple]: CONFIRMED
## [21380] Active tracks: #7 [apple]: CONFIRMED
## [21505] Active tracks: #7 [apple]: CONFIRMED
## [21631] Active tracks: #7 [apple]: CONFIRMED
## [21756] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [21881] Active tracks: #7 [apple]: CONFIRMED
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## [22131] Active tracks: #7 [apple]: CONFIRMED
## [22256] Active tracks: #7 [apple]: CONFIRMED
## [22381] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [22506] Active tracks: #7 [apple]: CONFIRMED
## [22632] Active tracks: #7 [apple]: CONFIRMED
## [22756] Active tracks: #7 [apple]: CONFIRMED
## [22882] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [23006] Active tracks: #7 [apple]: CONFIRMED
## [23132] Active tracks: #7 [apple]: CONFIRMED
## [23257] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [23381] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [23508] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [23631] Active tracks: #7 [apple]: TEMPORARILY_MISSING
## [23757] Active tracks: none
## [23881] Active tracks: none
## [24007] Active tracks: none
## [24133] Active tracks: none
## [24259] Active tracks: none
```

```
## [24382] Active tracks: none
## [24510] Active tracks: none
## [24634] Active tracks: none
## [24758] Active tracks: none
## [24882] Active tracks: none
## [25007] Active tracks: none
## [25132] Active tracks: none
## [25257] Active tracks: none
## [25382] Active tracks: none
## [25508] Active tracks: none
## [25633] Active tracks: none
## [25758] Active tracks: none
## [25883] Active tracks: none
## [26008] Active tracks: #9 [apple]: TENTATIVE
## [26134] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [26259] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [26383] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [26509] Active tracks: #9 [apple]: TEMPORARILY_MISSING
## [26633] Active tracks: none
## [26759] Active tracks: none
## [26884] Active tracks: #10 [knife]: TENTATIVE
## [27008] Active tracks: #10 [knife]: TEMPORARILY_MISSING | #11 [apple]: TENTATIVE
## [27135] Active tracks: #10 [knife]: TEMPORARILY_MISSING | #11 [apple]: CONFIRMED
## [27259] Active tracks: #10 [knife]: TEMPORARILY_MISSING | #11 [apple]: CONFIRMED
## [27384] Active tracks: #10 [knife]: TEMPORARILY_MISSING | #11 [apple]: CONFIRMED
## [27510] Active tracks: #10 [knife]: TENTATIVE | #11 [apple]: CONFIRMED | #12 [knife]: TENTATIVE
## [27634] Active tracks: #10 [knife]: TEMPORARILY_MISSING | #11 [apple]: TEMPORARILY_MISSING | #12 [kn
## [27759] Active tracks: #10 [knife]: TEMPORARILY_MISSING | #11 [apple]: CONFIRMED | #12 [knife]: TEMP
## [27884] Active tracks: #10 [knife]: TEMPORARILY_MISSING | #11 [apple]: CONFIRMED | #12 [knife]: TENT
## [28009] Active tracks: #10 [knife]: TEMPORARILY_MISSING | #11 [apple]: CONFIRMED | #12 [knife]: CONF
## [28134] Active tracks: #11 [apple]: CONFIRMED | #12 [knife]: CONFIRMED
## [28259] Active tracks: #11 [apple]: CONFIRMED | #12 [knife]: CONFIRMED
## [28384] Active tracks: #11 [apple]: CONFIRMED | #12 [knife]: TEMPORARILY_MISSING
## [28509] Active tracks: #11 [apple]: CONFIRMED | #12 [knife]: CONFIRMED
## [28634] Active tracks: #11 [apple]: TEMPORARILY_MISSING | #12 [knife]: CONFIRMED | #13 [knife]: TENT
## [28759] Active tracks: #11 [apple]: TEMPORARILY_MISSING | #12 [knife]: TEMPORARILY_MISSING | #13 [kn
## [28886] Active tracks: #11 [apple]: TEMPORARILY_MISSING | #12 [knife]: TEMPORARILY_MISSING | #13 [kn
## [29010] Active tracks: #11 [apple]: TEMPORARILY_MISSING | #12 [knife]: TEMPORARILY_MISSING | #13 [kn
## [29135] Active tracks: #12 [knife]: TEMPORARILY_MISSING | #13 [knife]: TENTATIVE
## [29260] Active tracks: #13 [knife]: CONFIRMED
## [29385] Active tracks: #13 [knife]: TEMPORARILY_MISSING
## [29510] Active tracks: #13 [knife]: TEMPORARILY_MISSING | #14 [knife]: TENTATIVE
## [29635] Active tracks: #13 [knife]: TEMPORARILY_MISSING | #14 [knife]: TEMPORARILY_MISSING
## [29761] Active tracks: #13 [knife]: TEMPORARILY_MISSING | #14 [knife]: TEMPORARILY_MISSING
## [29886] Active tracks: #14 [knife]: TEMPORARILY_MISSING
## [30011] Active tracks: #14 [knife]: TEMPORARILY_MISSING
## [30135] Active tracks: none
## [30261] Active tracks: none
## [30386] Active tracks: none
## [30511] Active tracks: none
## [30636] Active tracks: none
## [30761] Active tracks: none
## [30885] Active tracks: none
## [31011] Active tracks: none
```

```
## [31136] Active tracks: none
## [31261] Active tracks: none
## [31386] Active tracks: none
## [31512] Active tracks: none
## [31636] Active tracks: none
## [31761] Active tracks: none
## [31887] Active tracks: none
## [32012] Active tracks: none
## [32137] Active tracks: none
## [32262] Active tracks: none
## [32387] Active tracks: none
## [32512] Active tracks: none
## [32637] Active tracks: none
## [32762] Active tracks: none
## [32887] Active tracks: none
## [33013] Active tracks: #15 [apple]: TENTATIVE
## [33137] Active tracks: #15 [apple]: CONFIRMED
## [33262] Active tracks: #15 [apple]: CONFIRMED
## [33387] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [33513] Active tracks: #15 [apple]: CONFIRMED
## [33638] Active tracks: #15 [apple]: CONFIRMED
## [33762] Active tracks: #15 [apple]: CONFIRMED
## [33888] Active tracks: #15 [apple]: CONFIRMED
## [34013] Active tracks: #15 [apple]: CONFIRMED
## [34138] Active tracks: #15 [apple]: CONFIRMED
## [34263] Active tracks: #15 [apple]: CONFIRMED
## [34388] Active tracks: #15 [apple]: CONFIRMED
## [34513] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [34639] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [34763] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [34891] Active tracks: #15 [apple]: TEMPORARILY_MISSING
## [35013] Active tracks: none
## [35138] Active tracks: none
## [35263] Active tracks: none
## [35389] Active tracks: none
## [35514] Active tracks: none
## [35639] Active tracks: none
## [35765] Active tracks: none
## [35890] Active tracks: none
## [36016] Active tracks: none
## [36139] Active tracks: none
## [36265] Active tracks: none
## [36389] Active tracks: none
## [36514] Active tracks: none
## [36639] Active tracks: none
## [36764] Active tracks: none
## [36889] Active tracks: none
## [37016] Active tracks: none
## [37140] Active tracks: none
## [37265] Active tracks: none
## [37390] Active tracks: none
## [37515] Active tracks: none
## [37640] Active tracks: none
## [37765] Active tracks: none
```

```
## [37890] Active tracks: none
## [38016] Active tracks: none
## [38140] Active tracks: #16 [apple]: TENTATIVE
## [38267] Active tracks: #16 [apple]: CONFIRMED
## [38391] Active tracks: #16 [apple]: CONFIRMED
## [38517] Active tracks: #16 [apple]: TEMPORARILY_MISSING
## [38641] Active tracks: #16 [apple]: TEMPORARILY_MISSING
## [38767] Active tracks: #16 [apple]: TEMPORARILY_MISSING
## [38891] Active tracks: #16 [apple]: TEMPORARILY_MISSING
## [39017] Active tracks: none
## [39141] Active tracks: none
## [39266] Active tracks: none
## [39391] Active tracks: none
## [39517] Active tracks: none
## [39642] Active tracks: none
## [39766] Active tracks: #17 [knife]: TENTATIVE
## [39891] Active tracks: #17 [knife]: TEMPORARILY_MISSING
## [40017] Active tracks: #17 [knife]: TEMPORARILY_MISSING
## [40141] Active tracks: #17 [knife]: TEMPORARILY_MISSING
## [40266] Active tracks: #17 [knife]: TEMPORARILY_MISSING
## [40391] Active tracks: none
## [40517] Active tracks: none
## [40641] Active tracks: none
## [40767] Active tracks: none
## [40892] Active tracks: none
## [41018] Active tracks: none
## [41142] Active tracks: none
## [41267] Active tracks: none
## [41392] Active tracks: none
## [41517] Active tracks: none
## [41642] Active tracks: none
## [41768] Active tracks: none
## [41892] Active tracks: none
## [42017] Active tracks: none
## [42142] Active tracks: none
## [42268] Active tracks: #18 [apple]: TENTATIVE
## [42394] Active tracks: #18 [apple]: TEMPORARILY_MISSING | #19 [apple]: TENTATIVE
## [42518] Active tracks: #18 [apple]: TEMPORARILY_MISSING | #19 [apple]: TEMPORARILY_MISSING
## [42643] Active tracks: #18 [apple]: TENTATIVE | #19 [apple]: TEMPORARILY_MISSING
## [42768] Active tracks: #18 [apple]: TEMPORARILY_MISSING | #19 [apple]: TEMPORARILY_MISSING
## [42893] Active tracks: #18 [apple]: TEMPORARILY_MISSING | #19 [apple]: TEMPORARILY_MISSING
## [43018] Active tracks: #18 [apple]: TEMPORARILY_MISSING
## [43144] Active tracks: #18 [apple]: TEMPORARILY_MISSING
## [43269] Active tracks: none
## [43394] Active tracks: none
## [43518] Active tracks: none
## [43644] Active tracks: none
## [43769] Active tracks: none
## [43895] Active tracks: none
## [44018] Active tracks: none
## [44145] Active tracks: none
## [44270] Active tracks: none
## [44394] Active tracks: none
## [44520] Active tracks: none
```

```
## [44644] Active tracks: none
## [44770] Active tracks: none
## [44895] Active tracks: none
## [45019] Active tracks: none
## [45144] Active tracks: none
## [45269] Active tracks: none
## [45395] Active tracks: none
## [45519] Active tracks: #20 [apple]: TENTATIVE
## [45645] Active tracks: #20 [apple]: TEMPORARILY_MISSING
## [45772] Active tracks: #20 [apple]: TEMPORARILY_MISSING
## [45907] Active tracks: #20 [apple]: TENTATIVE
## [46021] Active tracks: #20 [apple]: TEMPORARILY_MISSING
## [46146] Active tracks: #20 [apple]: TEMPORARILY_MISSING
## [46270] Active tracks: #20 [apple]: TEMPORARILY_MISSING
## [46396] Active tracks: #20 [apple]: TEMPORARILY_MISSING
## [46520] Active tracks: none
## [46646] Active tracks: none
## [46771] Active tracks: none
## [46896] Active tracks: none
## [47022] Active tracks: none
## [47147] Active tracks: none
## [47272] Active tracks: none
## [47397] Active tracks: none
## [47522] Active tracks: none
## [47647] Active tracks: none
## [47772] Active tracks: #21 [knife]: TENTATIVE
## [47897] Active tracks: #21 [knife]: TEMPORARILY_MISSING
## [48022] Active tracks: #21 [knife]: TEMPORARILY_MISSING
## [48146] Active tracks: #21 [knife]: TEMPORARILY_MISSING
## [48272] Active tracks: #21 [knife]: TEMPORARILY_MISSING
## [48397] Active tracks: none
## [48522] Active tracks: none
## [48647] Active tracks: none
## [48772] Active tracks: none
## [48898] Active tracks: none
## [49023] Active tracks: none
## [49148] Active tracks: none
## [49273] Active tracks: #22 [apple]: TENTATIVE
## [49398] Active tracks: #22 [apple]: TEMPORARILY_MISSING | #23 [apple]: TENTATIVE
## [49522] Active tracks: #22 [apple]: TEMPORARILY_MISSING | #23 [apple]: TEMPORARILY_MISSING
## [49647] Active tracks: #22 [apple]: TEMPORARILY_MISSING | #23 [apple]: TEMPORARILY_MISSING
## [49773] Active tracks: #22 [apple]: TEMPORARILY_MISSING | #23 [apple]: TEMPORARILY_MISSING
## [49899] Active tracks: #23 [apple]: TEMPORARILY_MISSING
## [50023] Active tracks: none
## [50147] Active tracks: #24 [apple]: TENTATIVE
## [50273] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [50398] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [50523] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [50649] Active tracks: #24 [apple]: TEMPORARILY_MISSING
## [50773] Active tracks: #24 [orange (fruit)]: TENTATIVE
## [50898] Active tracks: #24 [orange (fruit)]: TEMPORARILY_MISSING
## [51023] Active tracks: #24 [orange (fruit)]: TEMPORARILY_MISSING
## [51149] Active tracks: #24 [orange (fruit)]: TEMPORARILY_MISSING
## [51273] Active tracks: #24 [orange (fruit)]: TEMPORARILY_MISSING
```

```
## [51398] Active tracks: none
## [51523] Active tracks: none
## [51650] Active tracks: none
## [51775] Active tracks: none
## [51899] Active tracks: #25 [apple]: TENTATIVE
## [52023] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [52150] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [52274] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [52399] Active tracks: #25 [apple]: TEMPORARILY_MISSING
## [52525] Active tracks: none
## [52650] Active tracks: none
## [52775] Active tracks: #26 [apple]: TENTATIVE
## [52899] Active tracks: #26 [apple]: TEMPORARILY_MISSING
## [53025] Active tracks: #26 [apple]: TEMPORARILY_MISSING
## [53150] Active tracks: #26 [apple]: TEMPORARILY_MISSING
## [53274] Active tracks: #26 [apple]: TENTATIVE
## [53400] Active tracks: #26 [apple]: CONFIRMED | #27 [knife]: TENTATIVE
## [53527] Active tracks: #26 [apple]: TEMPORARILY_MISSING | #27 [knife]: CONFIRMED
## [53651] Active tracks: #26 [apple]: TEMPORARILY_MISSING | #27 [knife]: TEMPORARILY_MISSING
## [53775] Active tracks: #26 [apple]: TEMPORARILY_MISSING | #27 [knife]: TEMPORARILY_MISSING
## [53900] Active tracks: #26 [apple]: TEMPORARILY_MISSING | #27 [knife]: TEMPORARILY_MISSING
## [54026] Active tracks: #27 [knife]: TEMPORARILY_MISSING
## [54150] Active tracks: none
## [54275] Active tracks: none
## [54400] Active tracks: none
## [54528] Active tracks: #28 [knife]: TENTATIVE
## [54652] Active tracks: #28 [knife]: TEMPORARILY_MISSING
## [54775] Active tracks: #28 [knife]: TEMPORARILY_MISSING
## [54900] Active tracks: #28 [knife]: TEMPORARILY_MISSING
## [55026] Active tracks: #28 [knife]: TEMPORARILY_MISSING
## [55153] Active tracks: none
## [55276] Active tracks: none
## [55401] Active tracks: none
## [55526] Active tracks: none
## [55654] Active tracks: none
## [55779] Active tracks: none
## [55901] Active tracks: none
## [56026] Active tracks: none
## [56151] Active tracks: none
## [56276] Active tracks: none
## [56401] Active tracks: none
## [56526] Active tracks: none
## [56651] Active tracks: #29 [apple]: TENTATIVE
## [56776] Active tracks: #29 [apple]: TEMPORARILY_MISSING
## [56901] Active tracks: #29 [apple]: TEMPORARILY_MISSING
## [57026] Active tracks: #29 [apple]: TEMPORARILY_MISSING
## [57152] Active tracks: #29 [apple]: TEMPORARILY_MISSING
## [57276] Active tracks: none
## [57401] Active tracks: none
## [57525] Active tracks: none
## [57651] Active tracks: none
## [57777] Active tracks: none
## [57902] Active tracks: #30 [knife]: TENTATIVE
## [58027] Active tracks: #30 [knife]: TEMPORARILY_MISSING
```

```
## [58152] Active tracks: #30 [knife]: TEMPORARILY_MISSING
## [58277] Active tracks: #30 [knife]: TEMPORARILY_MISSING
## [58402] Active tracks: #30 [knife]: TEMPORARILY_MISSING
## [58527] Active tracks: none
## [58653] Active tracks: none
## [58777] Active tracks: none
## [58903] Active tracks: none
## [59027] Active tracks: none
## [59153] Active tracks: none
## [59278] Active tracks: none
## [59403] Active tracks: none
## [59528] Active tracks: none
## [59653] Active tracks: none
## [59778] Active tracks: none
## [59903] Active tracks: none
## [60029] Active tracks: none
## [60153] Active tracks: #31 [apple]: TENTATIVE
## [60278] Active tracks: #31 [apple]: TEMPORARILY_MISSING
## [60404] Active tracks: #31 [apple]: TEMPORARILY_MISSING
## [60528] Active tracks: #31 [apple]: TEMPORARILY_MISSING
## [60653] Active tracks: #31 [apple]: TEMPORARILY_MISSING
## [60779] Active tracks: none
## [60904] Active tracks: none
## [61029] Active tracks: none
## [61154] Active tracks: none
## [61279] Active tracks: none
## [61404] Active tracks: none
## [61529] Active tracks: none
## [61654] Active tracks: none
## [61780] Active tracks: none
## [61905] Active tracks: none
## [62030] Active tracks: none
## [62155] Active tracks: none
## [62281] Active tracks: none
## [62404] Active tracks: none
## [62531] Active tracks: none
## [62655] Active tracks: none
## [62780] Active tracks: none
## [62906] Active tracks: none
## [63030] Active tracks: none
## [63155] Active tracks: none
## [63282] Active tracks: none
## [63406] Active tracks: none
## [63532] Active tracks: none
## [63656] Active tracks: none
## [63782] Active tracks: none
## [63905] Active tracks: none
## [64032] Active tracks: none
## [64156] Active tracks: none
## [64282] Active tracks: none
## [64406] Active tracks: none
## [64533] Active tracks: none
## [64656] Active tracks: none
## [64783] Active tracks: none
```

```
## [64907] Active tracks: none
## [65032] Active tracks: none
## [65157] Active tracks: none
## [65283] Active tracks: none
## [65406] Active tracks: none
## [65532] Active tracks: none
## [65656] Active tracks: none
## [65782] Active tracks: none
## [65907] Active tracks: none
## [66032] Active tracks: none
## [66158] Active tracks: none
## [66283] Active tracks: none
## [66407] Active tracks: none
## [66533] Active tracks: none
## [66658] Active tracks: none
## [66783] Active tracks: none
## [66908] Active tracks: none
## [67032] Active tracks: none
## [67158] Active tracks: none
## [67283] Active tracks: none
## [67408] Active tracks: none
## [67534] Active tracks: none
## [67659] Active tracks: none
## [67783] Active tracks: none
## [67908] Active tracks: none
## [68034] Active tracks: none
## [68159] Active tracks: none
## [68286] Active tracks: none
## [68408] Active tracks: none
## [68534] Active tracks: none
## [68659] Active tracks: none
## [68784] Active tracks: none
## [68909] Active tracks: none
## [69034] Active tracks: none
## [69161] Active tracks: none
## [69285] Active tracks: none
## [69411] Active tracks: none
## [69535] Active tracks: none
## [69661] Active tracks: none
## [69785] Active tracks: none
## [69910] Active tracks: none
## [70036] Active tracks: none
## [70160] Active tracks: none
## [70285] Active tracks: none
## [70410] Active tracks: none
## [70535] Active tracks: none
## [70660] Active tracks: none
## [70786] Active tracks: none
## [70911] Active tracks: none
## [71036] Active tracks: none
```

Data exploration: qualitative analysis

Here's what merely examining the output data tells us:

`categoryHistory` recording seemingly broken in the dataformat Looking at the first 5 lines and last 10 lines of the contents of `t2` we printed out: Specifically where `"record_type":"frame"`: for each track in `tracker_state.tracks` `categoryHistory` always seems to be `{}`. To confirm that the data is indeed broken we wrote the `print_category_history_values` to iterate over each measurement result and print out all unique category history values. We'll run it on all of our data now:

```
print_category_history_values(t1)
```

```
## Here are all the different categoryHistory values in tracks in frame-records in t1:
##
## {}
```

```
print_category_history_values(t2)
```

```
## Here are all the different categoryHistory values in tracks in frame-records in t2:
##
## {}
```

```
print_category_history_values(t3)
```

```
## Here are all the different categoryHistory values in tracks in frame-records in t3:
##
## {}
```

```
print_category_history_values(t4)
```

```
## Here are all the different categoryHistory values in tracks in frame-records in t4:
##
## {}
```

```
print_category_history_values(t5)
```

```
## Here are all the different categoryHistory values in tracks in frame-records in t5:
##
## {}
```

```
print_category_history_values(t6)
```

```
## Here are all the different categoryHistory values in tracks in frame-records in t6:
##
## {}
```

```
print_category_history_values(t7)
```

```
## Here are all the different categoryHistory values in tracks in frame-records in t7:
##
## {}
```

This confirms that category history saving in our data is broken. This warrants further investigation.

Rapid movement problematic for tracking This is clearly seen since we conduct measurements which differ from each other in movement speed and also performance. In `t1` we have a single stationary object. In `t3` we have a single slowly moving object and in `t4` we have a single repidly moving object. By looking at the timeline of the last 20 frames for each of these measurements we see:

```
print_track_timeline(tail(t1, 20))
```

```
## [1729537] Active tracks: #1 [bowl]: CONFIRMED
## [1729662] Active tracks: #1 [bowl]: CONFIRMED
## [1729788] Active tracks: #1 [bowl]: CONFIRMED
```

```
## [1729913] Active tracks: #1 [bowl]: CONFIRMED
## [1730038] Active tracks: #1 [bowl]: CONFIRMED
## [1730163] Active tracks: #1 [bowl]: CONFIRMED
## [1730288] Active tracks: #1 [bowl]: CONFIRMED
## [1730413] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1730540] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1730665] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1730789] Active tracks: #1 [bowl]: TEMPORARILY_MISSING
## [1730915] Active tracks: none
## [1731039] Active tracks: none
## [1731166] Active tracks: none
## [1731291] Active tracks: #2 [bowl]: TENTATIVE
## [1731414] Active tracks: #2 [bowl]: TEMPORARILY_MISSING
## [1731539] Active tracks: #2 [bowl]: TEMPORARILY_MISSING
## [1731664] Active tracks: #2 [bowl]: TENTATIVE
## [1731790] Active tracks: #2 [bowl]: TEMPORARILY_MISSING
## [1731914] Active tracks: #2 [bowl]: TEMPORARILY_MISSING
```

```
print_track_timeline(tail(t3, 20))
```

```
## [486832] Active tracks: #16 [apple]: TEMPORARILY_MISSING
## [486957] Active tracks: #16 [apple]: CONFIRMED
## [487083] Active tracks: #16 [apple]: CONFIRMED
## [487208] Active tracks: #16 [apple]: CONFIRMED
## [487333] Active tracks: #16 [apple]: CONFIRMED
## [487457] Active tracks: #16 [apple]: CONFIRMED
## [487582] Active tracks: #16 [orange (fruit)]: CONFIRMED
## [487708] Active tracks: #16 [orange (fruit)]: CONFIRMED
## [487831] Active tracks: #16 [orange (fruit)]: TEMPORARILY_MISSING
## [487958] Active tracks: #16 [orange (fruit)]: TEMPORARILY_MISSING
## [488082] Active tracks: #16 [orange (fruit)]: TEMPORARILY_MISSING
## [488208] Active tracks: #16 [orange (fruit)]: TEMPORARILY_MISSING
## [488332] Active tracks: none
## [488458] Active tracks: none
## [488583] Active tracks: none
## [488708] Active tracks: none
## [488832] Active tracks: none
## [488959] Active tracks: none
## [489083] Active tracks: none
## [489209] Active tracks: none
```

```
print_track_timeline(tail(t4, 20))
```

```
## [66715] Active tracks: #52 [bowl]: CONFIRMED
## [66839] Active tracks: #52 [bowl]: TEMPORARILY_MISSING
## [66964] Active tracks: #52 [bowl]: TEMPORARILY_MISSING
## [67091] Active tracks: #52 [bowl]: TEMPORARILY_MISSING
## [67214] Active tracks: #52 [bowl]: TEMPORARILY_MISSING
## [67339] Active tracks: none
## [67464] Active tracks: none
## [67590] Active tracks: none
## [67714] Active tracks: none
## [67840] Active tracks: none
## [67966] Active tracks: none
## [68091] Active tracks: none
```

```
## [68217] Active tracks: none
## [68345] Active tracks: none
## [68469] Active tracks: none
## [68594] Active tracks: none
## [68716] Active tracks: none
## [68841] Active tracks: none
## [68965] Active tracks: none
## [69091] Active tracks: none
```

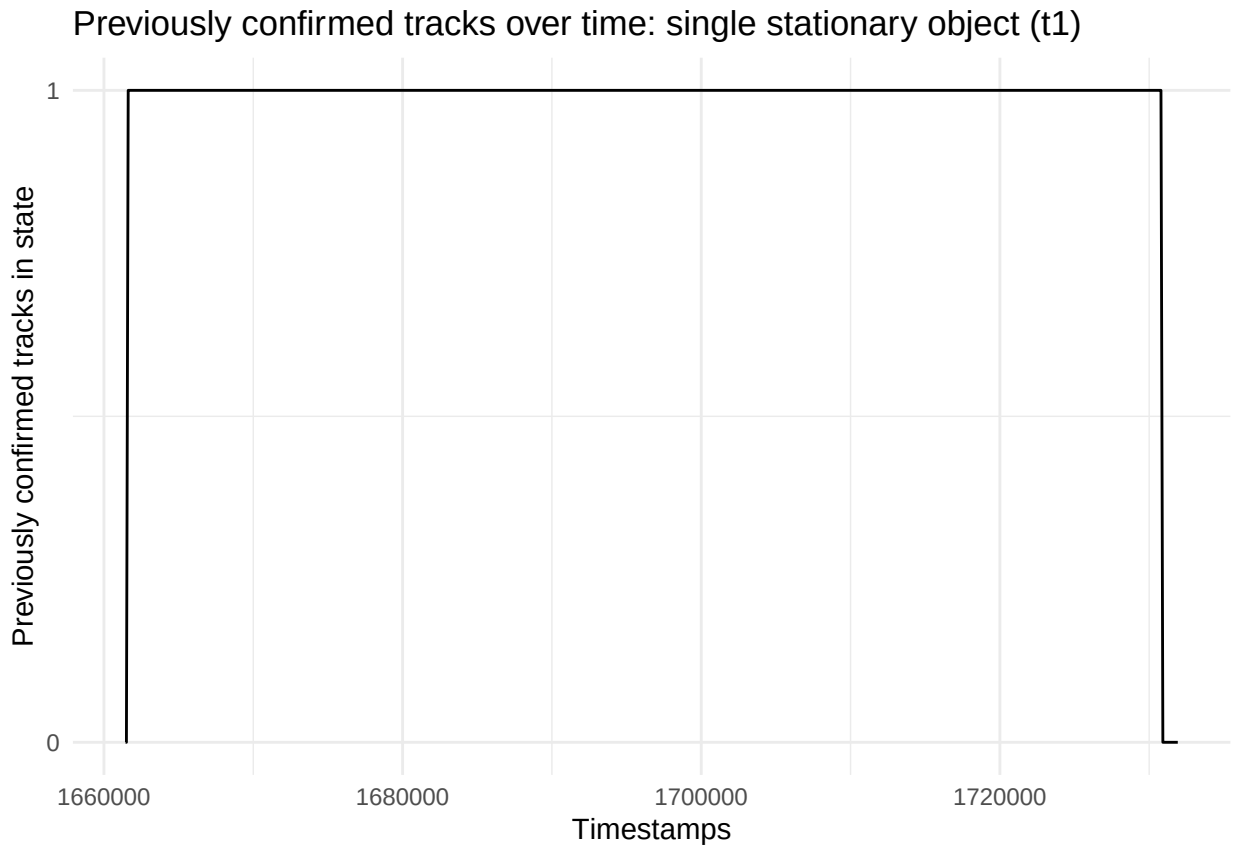
In `t1` only two objects are tracked over the experiment lifetime. In `t3` 16 objects are tracked and in `t4` 52 objects are tracked. So by increasing movement speed of the tracked objects generated tracks seemingly increase which points in the direction of a worsening tracking quality on increasing speed.

Since `t1`, `t3` and `t4` are all single-item experiments, the tracker works optimally when there is exactly one track in the state. Otherwise:

- Less than one track means that the tracker doesn't track enough things (likely due to not receiving enough consecutive detections)
- More than one track means that the tracker tracks too many things (likely due to ghost detections, location prediction failure causing track fragmentations or unsuccesful flip-flop protections causing track fragmentation)

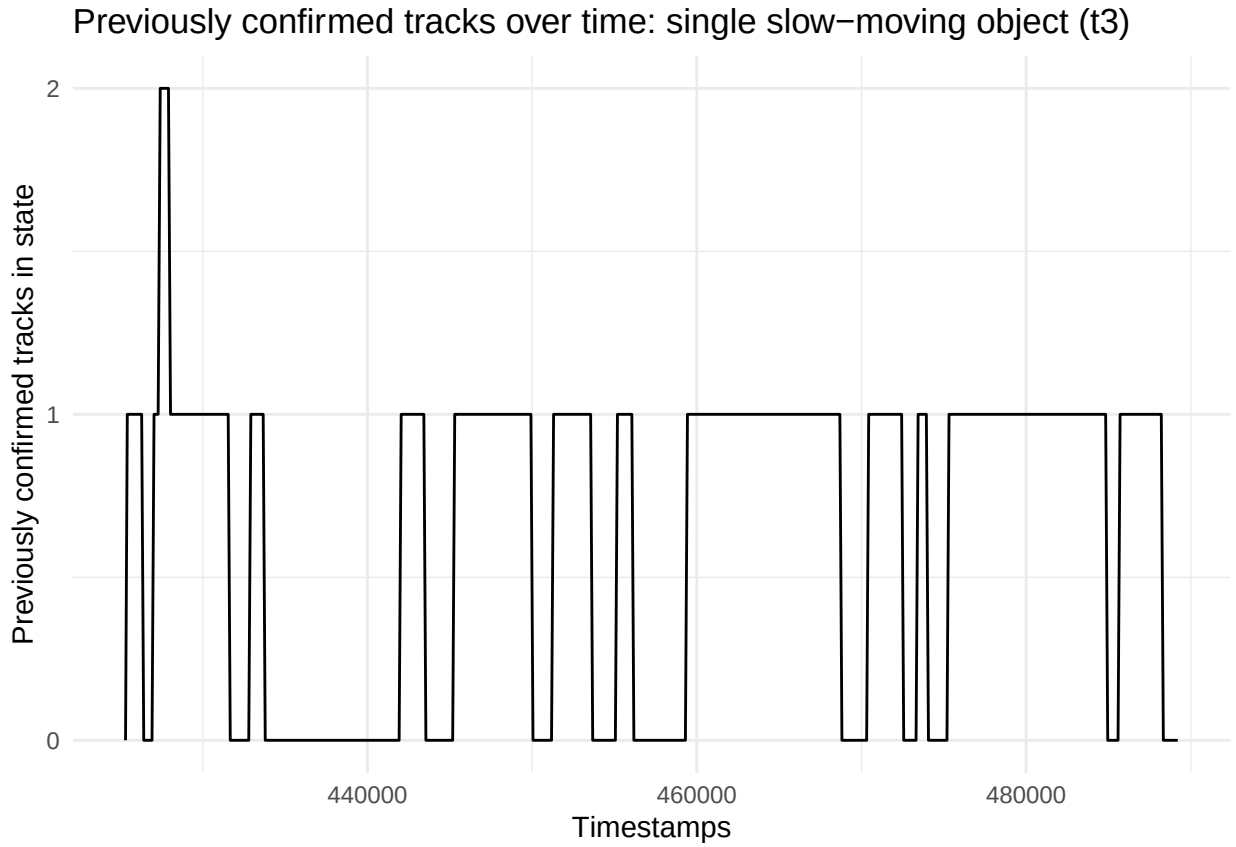
Here we graphically represent the amount of tracks in the status we have confirmed over the three different measurents

```
ggplot(confirmed_tracks_over_time(t1), aes(x = timestamp_ms,
                        y = confirmed_tracks)) +
  geom_line() +
  scale_y_continuous(breaks = scales::breaks_width(1)) +
  labs(
    title = "Previously confirmed tracks over time: single stationary object (t1)",
    x = "Timestamps",
    y = "Previously confirmed tracks in state"
  ) +
  theme_minimal()
```



So here as one can see tracking is *practically perfect*. At no time does the tracking break and at no point does the tracker track too many objects.

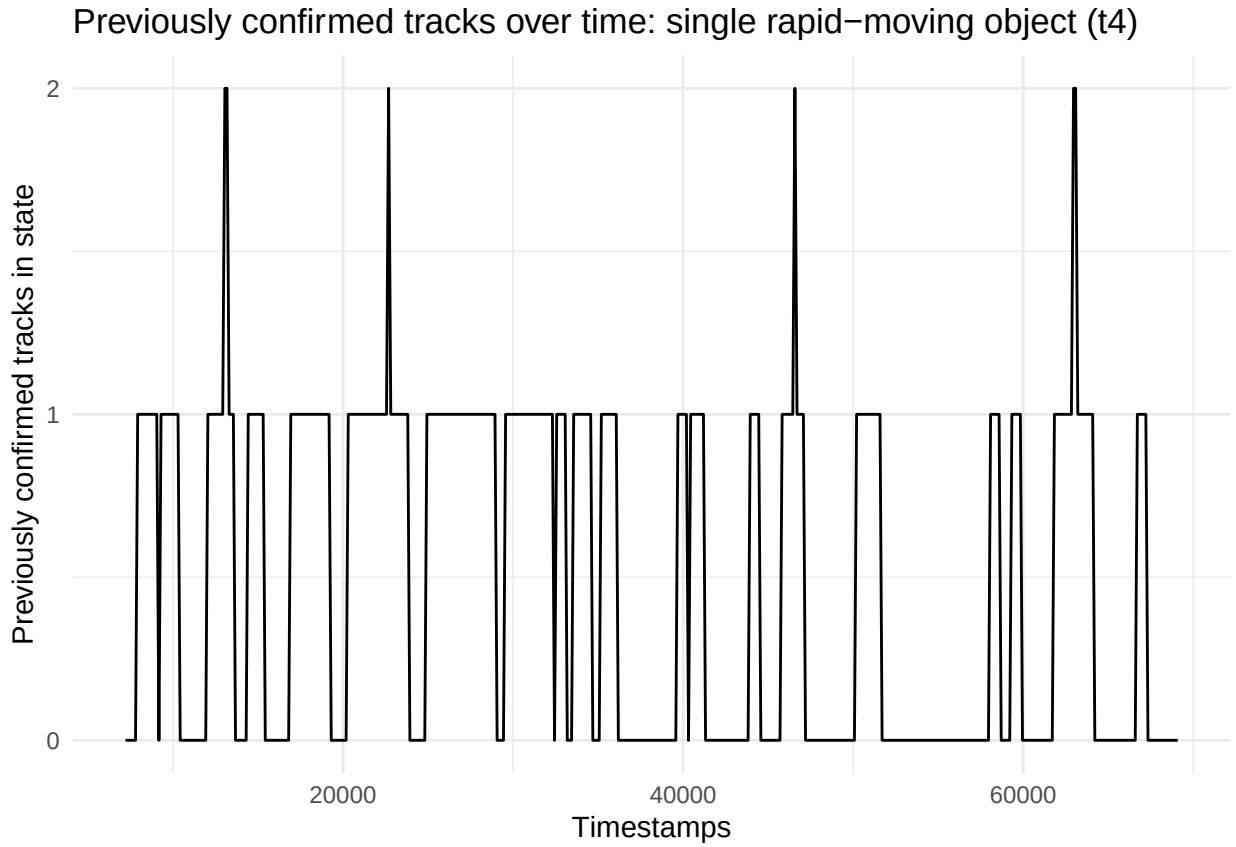
```
ggplot(confirmed_tracks_over_time(t3), aes(x = timestamp_ms,
                         y = confirmed_tracks)) +
  geom_line() +
  scale_y_continuous(breaks = scales::breaks_width(1)) +
  labs(
    title = "Previously confirmed tracks over time: single slow-moving object (t3)",
    x = "Timestamps",
    y = "Previously confirmed tracks in state"
  ) +
  theme_minimal()
```



Here we see:

- A few instances of track loss
- On instance of more than one simultaneous track (early on, before 430000)
- The big track loss around 440000 is particularly large. Probably due to bad detections or environmental complications rather than mere motion-related track loss

```
ggplot(confirmed_tracks_over_time(t4), aes(x = timestamp_ms,
                         y = confirmed_tracks)) +
  geom_line() +
  scale_y_continuous(breaks = scales::breaks_width(1)) +
  labs(
    title = "Previously confirmed tracks over time: single rapid-moving object (t4)",
    x = "Timestamps",
    y = "Previously confirmed tracks in state"
  ) +
  theme_minimal()
```



Here we see:

- Many instances of track loss. A lot more frequent than in `t3`. Likely due to increased speed. So speed increase would seem to correspond to track loss frequency
- We four cases of double tracking. Likely caused by unsuccesful category-flip-guarding causing double tracking (these are also notably small durations, which supports this theory!!) These too seems to increase in frequency as speed grows.

I also want to highlight instances with **seemingly obvious instances of rapid speed breaking tracking**

1. instance: measurement 4, timestamps 18433 to 19308 (inclusive)

```
print_track_timeline(t4[92:99])
```

```
## [18433] Active tracks: #10 [bowl]: CONFIRMED
## [18559] Active tracks: #10 [bowl]: TEMPORARILY_MISSING
## [18682] Active tracks: #10 [bowl]: CONFIRMED
## [18807] Active tracks: #10 [bowl]: TEMPORARILY_MISSING
## [18931] Active tracks: #10 [bowl]: TEMPORARILY_MISSING | #11 [bowl]: TENTATIVE
## [19057] Active tracks: #10 [bowl]: TEMPORARILY_MISSING | #11 [bowl]: TEMPORARILY_MISSING
## [19182] Active tracks: #10 [bowl]: TEMPORARILY_MISSING | #11 [bowl]: TEMPORARILY_MISSING
## [19308] Active tracks: #11 [bowl]: TEMPORARILY_MISSING
```

#10 and *#11* are likely the same object:

- both detected as bowls
- there is only one real-life object we attempt to track in `t4`
- here's how the timeline likely went:
    1. Tracker gets new location for object *#10* at 18682. New velocity calculation made, new location

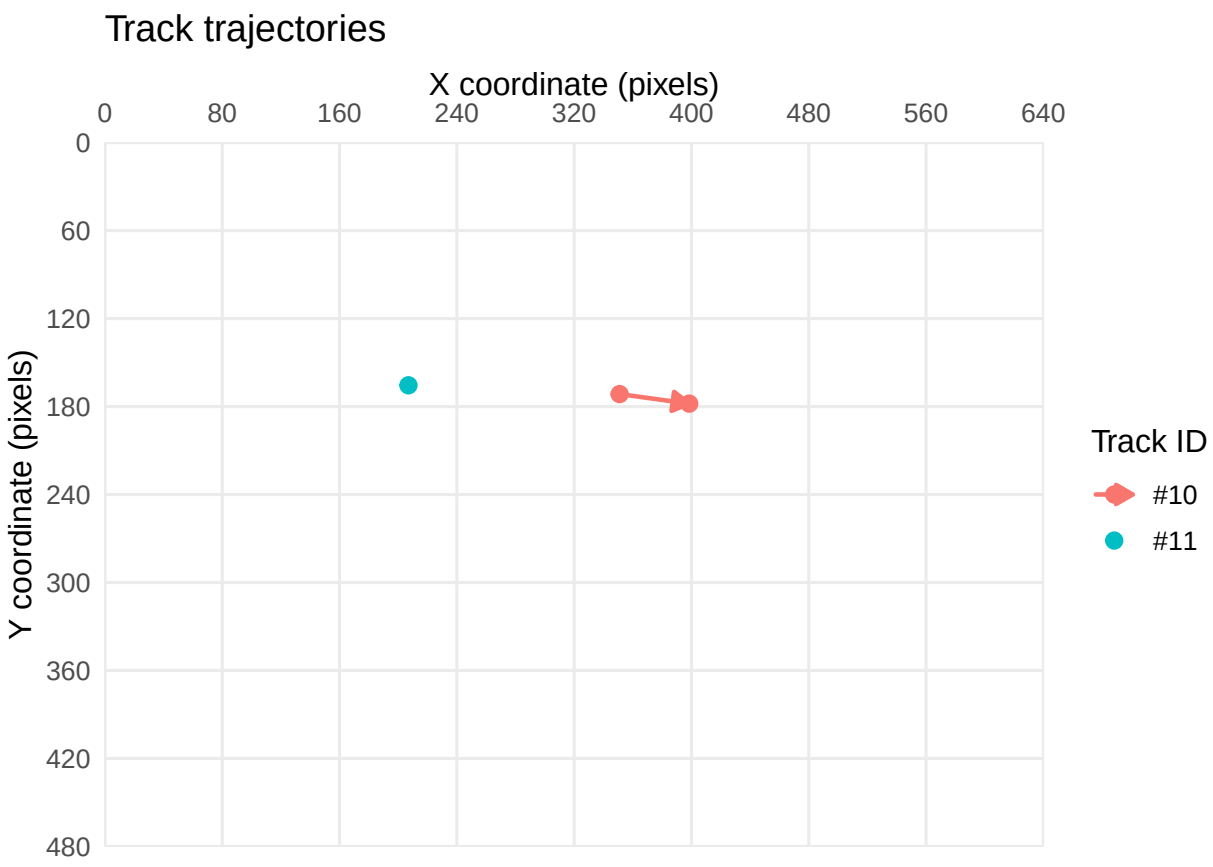
predicition made.
2. No detections at 18807, therefore *#10* marked `TEMPORARILY_MISSING`
3. New detection at 18931, doesn't match `LOOSE_MATCH_DISTANCE_PX`, (possibly not even `BACKUP_MATCH_DISTANCE_PX`), therefore it is marked as a new track even though it matches the category of *#10*

Us not being able to match the new detection at 18931 to the prediction produced at 18682 is a consequence of the following things:

- the missing frame at 18807
  - which if we assume this movement happens at a large velocity itself points to the hypothesis that the tracker is less reliable when the speed increases (due to worse detections)
- an inaccurate prediction
  - due to us not being able to update location & velocity estimations at 18807
  - or maybe the matching distances are too tight for a high velocity
  - or maybe acceleration/slowdown makes the predictions themselves wrong.

But to see how true our hypothesis is by looking at the trajectories of these tracks:

```
plot_track_trajectories(t4[92:99])
```



So here the hypothesis that only rapid speed broke the tracking can be discarded, since *#11* does not correspond to the direction *#10* was moving. Since *#11* is only detected for a single frame, this is likely a ghost.

2. instance: measurement 4, timestamps 35443 to 36193 (inclusive)

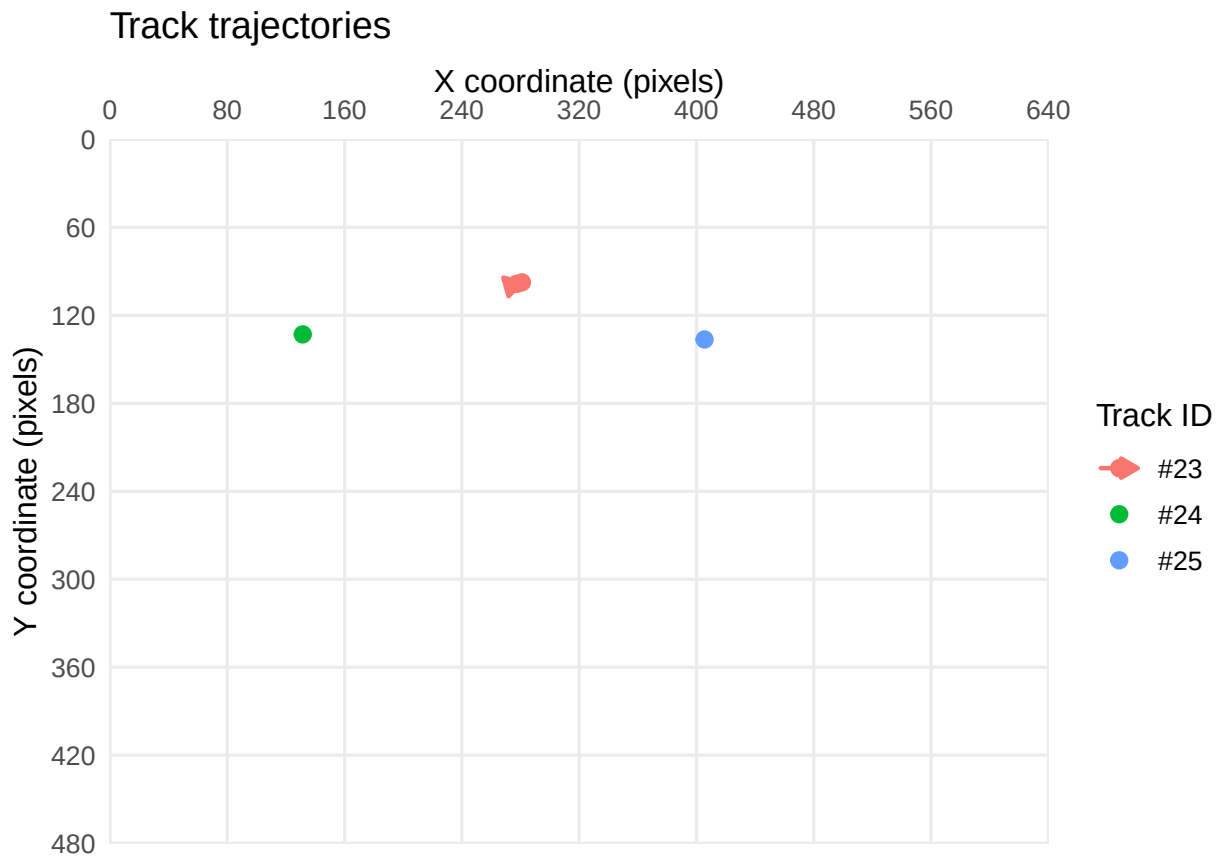
```
print_track_timeline(t4[228:234])
```

```
## [35443] Active tracks: #23 [bowl]: CONFIRMED
```

```
## [35568] Active tracks: #23 [bowl]: CONFIRMED
## [35693] Active tracks: #23 [bowl]: TEMPORARILY_MISSING
## [35820] Active tracks: #23 [bowl]: TEMPORARILY_MISSING | #24 [bowl]: TENTATIVE
## [35946] Active tracks: #23 [bowl]: TEMPORARILY_MISSING | #24 [bowl]: TEMPORARILY_MISSING
## [36070] Active tracks: #23 [bowl]: TEMPORARILY_MISSING | #24 [bowl]: TEMPORARILY_MISSING | #25 [bowl]
## [36193] Active tracks: #24 [bowl]: TEMPORARILY_MISSING | #25 [bowl]: TEMPORARILY_MISSING
```

By the same logic as above (all items are bowls, only one real life item is being moved in front of the camera), we make the same assumption here too (*#23*, *#24* and *#25* are the same item, which is given three different `instanceID`s due to the real world item "outrunning" the tracker's loose/backup matching treshold). Once again let us draw the trajectories of the tracked items in this pre-specified timeframe:

```
plot_track_trajectories(t4[228:234])
```



It is highly unlikely that all three tracks *#23*, *#24* and *#25* would be the same object, since this would require notable accelerations and direction changes in a few frames' timespan. However, notably *#25* roughly matches the direction *#23* was travelling towards so this may or may not be an actual case of a failed match (note: *#23* is still being tracked when *#25* appears) but at least *#24* is a ghost making these instances too not so obviously a case of tracking broken by rapid movement speed.

Both highlighted "seemingly obvious examples" of tracking being broken due to rapid speeds actually seem to be examples of ghost tracks breaking tracking instead. Still, I believe 52 total tracks on rapid speed vs 16 on slow speed vs 2 on standing still is an indicator that rapid speed is an issue. Lastly I want to look at the following part of `t4` where **tracking seems to break due to flip-flop handling failing:

```
print_track_timeline(t4[177:185])
```

```
## [29065] Active tracks: none
## [29191] Active tracks: #18 [bowl]: TENTATIVE
```

```
## [29313] Active tracks: #18 [bowl]: TEMPORARILY_MISSING | #19 [knife]: TENTATIVE
## [29438] Active tracks: #18 [bowl]: TENTATIVE | #19 [knife]: TEMPORARILY_MISSING
## [29563] Active tracks: #18 [bowl]: CONFIRMED | #19 [knife]: TEMPORARILY_MISSING
## [29690] Active tracks: #18 [bowl]: CONFIRMED | #19 [knife]: TEMPORARILY_MISSING
## [29814] Active tracks: #18 [bowl]: CONFIRMED | #19 [knife]: TEMPORARILY_MISSING
## [29939] Active tracks: #18 [bowl]: TEMPORARILY_MISSING
## [30063] Active tracks: #18 [bowl]: CONFIRMED
```

Here: Assuming that *#18* and *#19* is the same item:

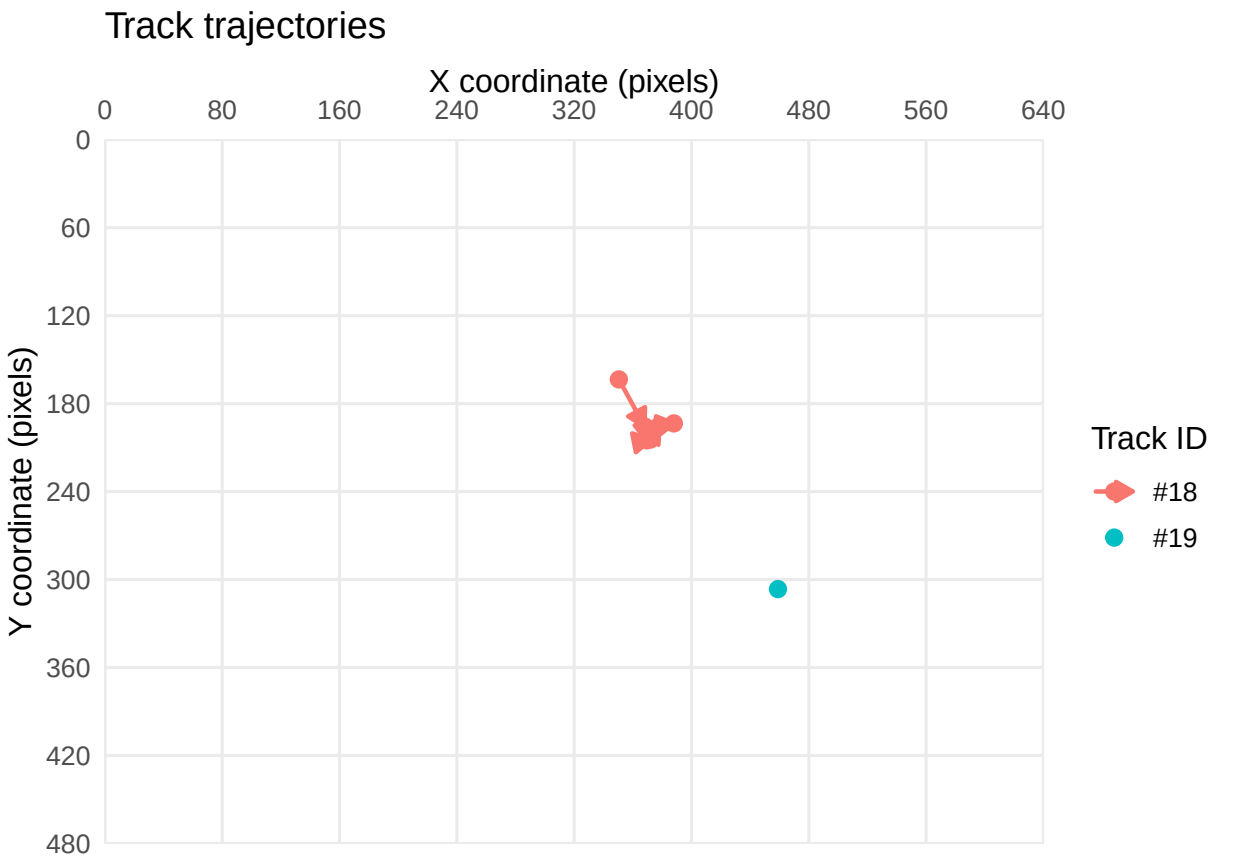
- It first gets tracked on `t4[178]`, getting the instanceID *#18* and being recognized as a *bowl*
- On `t4[179]` we see a new knife given an instanceID *#19* while *#18* goes missing
- On `t4[180]` *#18* (bowl) reappears and *#19* (knife) disappears
- *#19* (knife) is only ever seen on one frame, and *#18* gets confirmed and properly tracked after the prolonged confirmation period caused by it going missing on frame `t4[179]`

If *#18* and *#19* truely are the same item:

- `t4[179]` is a flip-flop (an item previously detected as a bowl now detected as a knife)
- The instanceID *#19* is given due to the detection at `t4[179]` falling outside of `LOOSE_MATCH_DISTANCE_PX` (possibly even `BACKUP_MATCH_DISTANCE_PX`)
- At `t4[179]` *#18* has only ever been detected one time meaning that we have no basis for velocity calculation & location prediction due to a lack of history & we only have a single known category from a single known detection; This situation is hard to guard against even if we confirm this is a case of flip-flopping due to rapid speed

Let's look at the actual trajectories at `t4[177:185]`:

```
plot_track_trajectories(t4[177:185])
```



Okay soo looking at the evidence; Once again we seem to see a ghost detection due to how inconsistant the position of *#19* is compered to the rest of the data points of *#18*. Considering that the *#19* detection comes between the first and the second *#18* detection, we'd expect any flip-flop to be at around $(360, 180)$. So all our concrete examples seem to be ghost detections rather than obvious examples of rapid speed breaking tracking (flip-flop protection or tracking logic breaking), but still the 52 vs 16 vs 2 total tracks clearly shows that the system does have issues with rapid speeds. It's just, we seemingly see more ghosts at rapid speeds rather than some logical component not working.